
A Unified Convergence Theory for Non-Stationary Reinforcement Learning

Anonymous Author(s)

Affiliation

Address

email

Abstract

Non-stationary reinforcement learning lacks a unified convergence theory: variation-budget regret bounds, sliding-window methods, tempo-and-forgetting analyses, two-term decompositions, and causal-RL frameworks each address fragments without composing. We present a unified theory through the composition of four structural elements: (i) a two-gap diagnostic separating goal-feasibility from policy-quality; (ii) a point-mass reverse-KL/TV identity under deterministic optimum, $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}}$, an *exact* algebraic identity that lies strictly below the Pinsker and Bretagnolle–Huber envelopes at this deterministic corner (collapsing the BH inequality’s envelope to a single value rather than introducing a new bounding technique); (iii) a multi-factor strategic tempo $\mathcal{T}_\Sigma = \min_{(i,j)} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$ with a structural forgetting prerequisite $\overline{\mathcal{T}}_\Sigma > \rho_\Sigma / R_\Sigma$ as a survival inequality; (iv) closed-loop interventional access making regret bounds learnable from on-policy data via the Pearl causal hierarchy. Composing these gives a cumulative dynamic regret theorem with rate $O(V_{\max} \sqrt{(B_T + 1) T})$ where B_T counts optimum-change events. We further establish a structural failure class: every gain-decay update (effective per-element gain $\alpha_{ij}^{(t)} \rightarrow 0$, including count-accumulating Bayesian schemes and Robbins–Monro gradient methods) eventually violates the forgetting prerequisite for any positive disturbance rate, while finite-gain mechanisms admit bidirectional thresholds (constant-step stochastic approximation, sliding windows, bounded memory, block restart). The point-mass identity is coordinate-optimal among bounds depending only on total variation, chain-rule compositional to multi-step trajectory regret as behavior-cloning loss against the optimal trajectory, and load-bearing for the cumulative theorem. The deterministic-optimum scope extends perturbatively to ϵ -stochastic and softmax-regularized policies with $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ corrections respectively (exponentially small with polynomial prefactor). Theory-only; Lemma 5.2 makes the ProST reduction rigorous at the sector-parameter level, while their published experiments motivate the strategic-tempo component.

1 Introduction

Non-stationary reinforcement learning has matured along four largely separate research tracks. Each has produced rigorous results; none has been composed with the others into a single convergence theory.

The four tracks are visible in the recent literature as parallel lineages. **Variation-budget dynamic regret under drift**^[1–4] gives sublinear dynamic regret under bounded total variation in reward and transition dynamics. **Two-term regret decompositions**^[5–7] split the dynamic regret along an *exploration-vs-adaptation* axis, isolating the cost of confidence-set construction from the cost of

tracking a moving optimum. **Tempo and forgetting analyses**^[8–12] make policy-update timing or evidence-discount rate an explicit convergence variable. **Causal and interventional access**^[13–18] uses causal-graph structure to sharpen sample complexity, but operates in stationary settings only.

The information-theoretic regret literature^[19–24] traverses these tracks at a different layer, using entropy, mutual information, information ratio, Pinsker, or Hellinger to bound regret. Notably absent from that literature, as we document below (Section F), is an *exact* reverse-KL / total-variation identity at the deterministic-optimum corner: a sharp action-distribution regret coordinate that strictly improves both Pinsker and the Bretagnolle–Huber inequality^[25] at this point. The Bretagnolle–Huber inequality is itself absent from the retrieved RL/non-stationary corpus *as an upper-bound regret coordinate* (its use in lower-bound proofs via Le Cam’s method^[26,27] is well established and not contested here); the *exact identity at its point-mass corner* — which we establish — is *a fortiori* absent.

These four tracks share an evident common ancestor — the dynamic-regret analysis of online MDPs — but no published framework composes them. In particular, no framework combines (a) a regret decomposition along the *goal-feasibility-vs-policy-quality* axis (distinct from the exploration-vs-adaptation decompositions); (b) an exact point-mass reverse-KL/TV regret identity under deterministic optimum, strictly tighter than the Bretagnolle–Huber inequality at this corner; (c) a structural survival inequality threading rate-of-policy-revision against environment-side disturbance; and (d) a closed-loop causal-access argument that makes the regret bound *learnable* on-policy rather than merely provable analytically. The fundamental question remains open:

Can we design a unified convergence theory for non-stationary reinforcement learning that handles non-stationarity, has explicit metric structure on policy space, and is learnable from on-policy data?

In this paper we present the first composition that delivers all three properties jointly. Composing four structural elements — a two-gap diagnostic, a point-mass reverse-KL/TV identity, a multi-factor strategic tempo with forgetting prerequisite, and closed-loop interventional access — yields a cumulative dynamic regret theorem with rate $\text{DynReg}(T) \leq \tilde{O}(N_h \sqrt{(B_T + 1)T})$ in the piecewise-stationary regime, where B_T counts optimum-change events along the realized trajectory and N_h is the planning horizon. The per-round coordinate of this rate is sharper than both Pinsker and Bretagnolle–Huber at the deterministic- π^* corner. The technical anchor is an exact algebraic identity:

$\boxed{\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)}},$ which holds for any policy Q with $Q(a^*) > 0$ and lies *strictly below* both the Pinsker envelope $\sqrt{D_{\text{KL}}/2}$ and the Bretagnolle–Huber envelope $\sqrt{1 - e^{-D_{\text{KL}}}}$ at this corner.

Roadmap Section 2 situates the four-track landscape. Section 3 sets minimal preliminaries. Section 4 presents the four-component composition theorem with narrative unpacking and a defense of why each component is necessary. Section 5 sketches the proof via four key lemmas chained through a block decomposition. Section 6 gives concluding observations and practitioner takeaways. Full proofs and auxiliary mathematical material are in the appendices.

1.1 Contribution

The contribution has the shape — recognized by the NeurIPS Theory Track guidelines as a valid form of originality — of “*a novel combination of existing techniques [where] the reasoning behind this combination is well-articulated.*” Three specific moves carry the paper.

(i) Point-mass reverse-KL/TV regret identity (Section 4, full development Section 5.1). Under deterministic optimum policy $\pi^* = \delta_{a^*}$, an elementary calculation gives the *exact identity* $D_{\text{KL}}(\pi^* \| Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q))$, equivalently $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$. Substituting this into the classical Bretagnolle–Huber inequality $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$ yields $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, which holds *strictly* on $D_{\text{KL}} > 0$ since $x < \sqrt{x}$ on $(0, 1)$. At the deterministic- π^* corner the identity supplies the *exact* TV value strictly below the BH envelope, positioning BH as the lineage the identity sits within rather than as the source it instantiates. Composed with the textbook total-variation regret bound (full development Section 5.1), the identity yields a tight two-sided characterization $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)}) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)})$, *strictly below* the Pinsker form $V_{\max} \sqrt{D_{\text{KL}}/2}$ — at $D_{\text{KL}} = 0.01$ the identity bound is $7\times$ tighter; at

88 $D_{\text{KL}} > 2$ Pinsker becomes vacuous against the trivial V_{max} envelope while the identity continues to
 89 saturate at V_{max} . The deterministic- π^* scope is not a hard wall: the identity extends *perturbatively* to
 90 ϵ -greedy and softmax-regularized optima with quantified $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\text{min}}/\tau))$
 91 corrections respectively (full development in Section 5.1 and Section B.2), recovering exact equality
 92 at $\epsilon = 0 / \tau \rightarrow 0$.

93 **(ii) Strategic tempo with forgetting prerequisite (Section 4, full development Section 5.2).** A
 94 multi-factor strategic tempo $\mathcal{T}_\Sigma$ — composed per revisable policy element of an observation rate
 95 ν , an identifiability coefficient ι , and an update gain η — yields a structural survival inequality
 96 $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$ in which all three factors are multiplicatively
 97 load-bearing per element, with the *bottleneck* (per-element minimum) as aggregator. The familiar
 98 single-factor form $(1 - \lambda) > \rho_\Sigma / R_\Sigma$ is recovered as a corollary under uniform-Regime-A Beta-
 99 Bernoulli normalization. A structural-class theorem identifies the *gain-decay* class $\mathcal{A}_{\text{decay}}$ (count-
 100 accumulating Bayesian updates, observation-aggregating schemes without restart, gradient-based
 101 Robbins–Monro decaying-step methods) as one in which every agent eventually violates the threshold
 102 for any positive disturbance rate, while non-accumulating mechanisms (constant-step SA, sliding-
 103 window, bounded-memory, block-restart) face *bidirectional* ceilings. This *lifts* the tempo result of^[8]
 104 from a single-factor hyperparameter-optimization claim to a multi-factor structural-threshold claim.
 105 The forgetting prerequisite is the strategic analog of the persistence condition in adaptive control^[28],
 106 with $(1 - \lambda)$ playing the role of adaptive tempo per element.

107 **(iii) Closed-loop interventional access (Section 4, full development Section 5.3).** With explicit
 108 grounding in Bareinboim’s causal-hierarchy theorem^[29], the agent in the feedback loop *generates*
 109 Pearl Level-2 (interventional) data by construction: the action a_t causally precedes o_{t+1} , so under
 110 positivity, sequential ignorability, and a known action mechanism, the loop’s data-generating process
 111 is interventional rather than associational. This makes the regret bound of (i) *learnable* from on-policy
 112 interaction in regimes with sufficient identifiability, not merely provable in principle. The argument is
 113 implicit in the action-perception-loop framing of active inference^[30,31] and the causal-RL line^[13–18];
 114 we make the Bareinboim-hierarchy connection explicit, partition usable strength of identification
 115 into three regimes (A: full; B: partial; C: observation-only), and flag honestly where the loop yields
 116 interventional *data* without yielding identified *do*-estimates.

117 **Connective tissue: the two-gap diagnostic (Section 4).** A separate decomposition of the regret
 118 signal — the *satisfaction gap* δ_{sat} (the goal is unmet under the best available one-step improvement)
 119 versus the *control regret* δ_{regret} (the gap between the best available improvement and the agent’s current
 120 policy) — provides the corrective-action routing that ties (i)–(iii) together. The 2×2 disambiguation
 121 routes four regimes (success, strategy problem, capability limit, both) to four distinct corrective
 122 actions, along an axis orthogonal to the exploration-vs-adaptation decompositions of^[5–7].

123 **Composition** The four together yield (Section 4) a non-stationarity-aware convergence theory with
 124 three properties no existing strand exhibits jointly: the bound *handles non-stationarity* (via tempo and
 125 forgetting); has *explicit metric structure on policy space* (via the point-mass reverse-KL identity); is
 126 *learnable from on-policy data* (via closed-loop interventional access). Composition with a variation-
 127 budget block argument under a restart-on-change base learner yields the cumulative dynamic regret
 128 bound stated above — sharper per-round coordinate than Pinsker/BH, with the cumulative rate sitting
 129 in the piecewise-stationary B_T family of^[1] (*not* directly comparable to the continuous-variation
 130 rate $\tilde{O}(SA V_T^{1/3} H T^{2/3})$ of^[3]; the two regimes coincide up to constants under abrupt-magnitude
 131 piecewise-stationarity). Each of the four components is itself a theorem (cited or proved); the paper’s
 132 contribution is the recognition that the four together close a story none of the strands closes individually,
 133 plus the cumulative-regret rate that follows from composing the per-round identity with a base-learner
 134 stationary rate.

135 1.2 Scope and limitations

136 **What “convergence” means here** We do not claim pointwise policy convergence — under genuine
 137 non-stationarity the optimum is itself moving. By *convergence* we mean: per-round regret-coordinate
 138 convergence within stationary blocks; cumulative dynamic regret with sublinear / vanishing-average
 139 rate when $B_T = o(T)$ (Cesàro tracking $\frac{1}{T} \sum_t (V_t^* - V(\pi_t)) \rightarrow 0$); and ultimate boundedness of
 140 the strategic mismatch $\|\delta_\Sigma\|$ under the tempo threshold. Pointwise $V(\pi_t) \rightarrow V^*$ is structurally
 141 unavailable under genuine non-stationarity; the Cesàro tracking statement is the right replacement.

Theory only No experiments. The point-mass identity is mathematically airtight (a two-line direct calculation that specializes the BH family at its deterministic- π^* corner). The empirical grounding for the strategic-tempo claim is provided indirectly through reduction to^[8] ProST as a special case (Section 5); at the steady-state sector-parameter level the reduction is rigorous.

Canonical scope The composition theorem holds under deterministic optimum policy, bounded value range, isolated optimum (so $\Delta_{\min} > 0$), and a singular causal trajectory in the sense made precise in Section 3. The deterministic- π^* scope extends *perturbatively* to ϵ -stochastic and softmax-regularized optima (Section 5.1); tied-optimum is a degraded one-sided form (Section C.5). Each axis of the canonical scope degrades cleanly outside it (we discuss the failure modes in Section 6).

Comparator regime The cumulative rate is in the piecewise-stationary B_T family, not the continuous-variation V_T family of^[3]. Generalizing the per-round-identity route to continuous variation is open (Section 6).

2 Related Work

The four tracks named in Section 1 organize the literature.

Variation-budget dynamic regret ^[1–4,32] establish dynamic regret bounds under a *variation budget* V_T on cumulative reward and transition variation.^[3] achieves the near-optimal continuous-variation rate $\tilde{O}(SA V_T^{1/3} H T^{2/3})$ via RestartQ-UCB;^[1] treats the piecewise-stationary case with restart-on-change. Our cumulative rate $\tilde{O}(N_h \sqrt{(B_T + 1)T})$ sits in the piecewise-stationary family — same block-shape as Cheung-style restart-on-change — recovered as a corollary of composition with the per-round identity. The novelty is the *threshold form* of the strategic-tempo inequality (environment regimes where no $\lambda \in (0, 1)$ stabilizes the sector-Lyapunov mismatch), which dynamic-regret-optimization analyses do not surface.

Two-term regret decompositions along the exploration-vs-adaptation axis ^[5–7] decompose the dynamic regret into a *confidence-set construction* term and an *adaptation-under-non-stationarity* term.^[33] treats a structurally adjacent question — constraint-region feasibility — using a two-term decomposition along that axis. Same shape, different axis: ours is goal-feasibility-vs-policy-quality (the satisfaction gap distinguishes “the goal is unattainable from M_t ” from “the current policy is suboptimal”); theirs is uncertainty-source. Neither subsumes the other: a Long-Fei Li-style exploration term can be present in any of our 2×2 cells, and an adaptation term likewise. The two axes are orthogonal.

Tempo and forgetting ^[8] (ProST) defines an agent tempo as the schedule of policy update times $\{t_1, \dots, t_K\}$ and computes the schedule that minimizes a dynamic regret upper bound.^[9] (Pausing Policy Learning) refines this with non-zero hold durations.^[10–12] establish forgetting / sliding-window analyses for non-stationary linear MDPs and bandits. We *lift* the ProST move along two axes simultaneously: single-factor $\rightarrow$ multi-factor (per-element $\nu \cdot \iota \cdot (1 - \lambda)$ rather than scalar update frequency); and hyperparameter-optimization $\rightarrow$ structural-survival inequality (the threshold below which no schedule persists, irrespective of which schedule is chosen). ProST recovers as the Beta-Bernoulli special case $|E| = 1, \nu = \iota = 1$.

Causal and interventional access ^[13–18,34] use causal-graph structure to sharpen sample complexity in RL. Their setting is *stationary*. We compose the closed-loop interventional-access argument with non-stationarity. The connection between Pearl’s causal hierarchy^[35] and the data-generating process of an agent in a feedback loop^[29] is the technical bridge.

Information-theoretic regret (cross-cutting) ^[19–24,36] traverse the above tracks at a different layer, using entropy / mutual information / information ratio / Pinsker / Hellinger as the divergence backbone. The Russo–Van Roy information ratio uses Pinsker as its Cauchy–Schwarz follow-up; the IDS line and its descendants use mutual-information bounds. The Bretagnolle–Huber inequality^[25] — well-known in lower-bound proofs via Le Cam’s method^[26,27] — does not appear *as an upper-bound regret coordinate* in this corpus, and its exact point-mass corner (our Lemma 5.1) is *a fortiori* absent. We document the retrieval scope and methodology in Section F.

191 **Adjacent (satisficing / feasibility)** ^[37,38] use *satisficing* to mean “any policy above acceptance
 192 level β is acceptable.” Our δ_{sat} is structurally distinct: $V_O^{\min}$ is a property of the objective (set by
 193 the application domain), and $\delta_{\text{sat}} > 0$ signals goal-relative *infeasibility* (no policy in Π meets the
 194 threshold), not policy-relative satisficing.

195 **Contemporaneous (post-March 2026)** ^[38,39]; cited and distinguished without empirical-
 196 comparison demand per the contemporaneous-work cutoff convention.

197 3 Preliminaries

198 We adopt strict minimalism: only the notation and assumptions used directly in the main result and
 199 the mechanism statements appear here. Convention-hierarchy details (one-step vs. receding-horizon
 200 vs. Bellman continuation), the action-gap matching argument, tied-optimum extensions, and the
 201 deterministic-trajectory deployment modes are deferred to Section C.

202 **Episodic non-stationary MDP** A finite-horizon non-stationary MDP $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$ with state
 203 space $\mathcal{S}$, finite action space $\mathcal{A}$, time-indexed transition kernels $P_t(\cdot \mid s, a)$ and reward functions
 204 $r_t(s, a)$ allowed to vary across rounds t , and finite planning horizon N_h . Standard non-stationary-RL
 205 boundedness assumptions^[1]: per-step reward in $[0, 1]$.

206 **Variation regimes** A variation budget V_T measures continuous variation in transitions and rewards.
 207 We work in the *piecewise-stationary* specialization: time partitions into $B_T + 1$ stationary intervals
 208 separated by optimum-change events, with $B_T := |\{t : a_t^* \neq a_{t-1}^*\}|$. The two regimes are distinct in
 209 general; under abrupt-magnitude- δ piecewise-stationarity they coincide up to constants ($V_T \asymp \delta B_T$).
 210 Our results are stated for B_T ; the continuous- V_T extension is open (Section 6).

211 **Policy distribution** The agent’s policy at round t is a distribution $Q_t(\cdot \mid s)$ over actions; we write Q
 212 when state and round are understood. We adopt the canonical scope: the optimum policy at (M_t, s)
 213 is *deterministic*, $\pi^*(\cdot \mid M_t, s) = \delta_{a^*(s)}$ where $a^*(s) := \arg \max_a Q^\pi(s, a)$ is the optimal action
 214 under the agent’s current model M_t . Deterministic π^* is canonical for finite-MDP RL with isolated
 215 optima^[27]; the perturbative extension to ϵ -stochastic and softmax-regularized optima (Section 5.1,
 216 full derivation Section B.2) covers smooth deviations.

217 **Value object** Given an objective O , model M_t , policy π , and horizon N_h : $Q_O(M_t, a; \pi_{\text{cont}}, N_h) =$
 218 $\mathbb{E}[V_O(\tau) \mid M_t, \text{do}(a_t = a), a_{t+1} : \sim \pi_{\text{cont}}]$. The $\text{do}(\cdot)$ ^[35] flags intervention on a_t , not conditioning.
 219 We adopt the **one-step improvement** convention $\pi_{\text{cont}} = \pi_{\text{current}}$ as default: most conservative, compa-
 220 rable across rounds, fixed-point-free. Receding-horizon and Bellman alternatives are in Section C.1.

221 **Bounded value range** $V_{\max}(M_t) := \max_a Q_O(M_t, a) - \min_a Q_O(M_t, a)$, finite under bounded
 222 reward and finite horizon.

223 **Action gap (isolated optimum)** $\Delta(a) := Q_O(a^*) - Q_O(a) \in [0, V_{\max}]$, $\Delta_{\min} := \min_{a \neq a^*} \Delta(a) >$
 224 0 whenever the optimum is isolated.

225 **Strategic regret** For policy distribution Q : $R(Q) := Q_O(a^*) - \mathbb{E}_{a \sim Q}[Q_O(a)] = \sum_{a \neq a^*} Q(a) \cdot$
 226 $\Delta(a)$. Three classical forms — $\mathbb{E}_{\pi^*}[V] - \mathbb{E}_Q[V]$, $V(a^*) - \mathbb{E}_Q[V]$, $\mathbb{E}_{\pi^*}[V - V_Q]$ — coincide under
 227 deterministic π^* .

228 **Total variation and reverse-KL** For finite action measures P, Q on $\mathcal{A}$, $\text{TV}(P, Q) :=$
 229 $\frac{1}{2} \sum_a |P(a) - Q(a)|$, $D_{\text{KL}}(P \parallel Q) := \sum_a P(a) \log \frac{P(a)}{Q(a)}$ with the convention $0 \log 0 = 0$.
 230 Our bounds use the *reverse* KL direction (with $P = \pi^*$); the forward direction $D_{\text{KL}}(Q \parallel \pi^*)$ is $+\infty$
 231 whenever Q has off-optimum mass and is therefore vacuous as a regret coordinate (full direction-
 232 forcing argument in Section C.3).

233 **Singular trajectory** The agent operates on a single, non-forkable causal trajectory: a_t causally
 234 precedes o_{t+1} , and the model state M_t ’s sufficiency is defined relative to *this* trajectory rather than to

a model-state equivalence class. This is the operational ground for the do-operator above and for the closed-loop interventional-access argument (Section 5.3).

4 Main Result

We present the four components first as motivation, then state the formal composition theorem, then unpack what its conclusions mean.

4.1 The four components

Each component supplies a *distinct epistemic property* that the others cannot deliver. The composition is necessary for the joint properties (rate + persistence + learnability + corrective-action routing), not for the rate alone. The component-by-component failure-mode analysis is in Section 4.4 below.

Component 1: Two-gap diagnostic — corrective-action routing Let Π denote the policy class available to the agent and let $V_O^{\min}$ denote the threshold trajectory value above which the objective is “met.” Define *objective attainability* $A_O(M_t; \Pi, N_h) := \sup_{\pi \in \Pi} V_O(M_t, \pi; N_h)$, the *satisfaction gap* $\delta_{\text{sat}} := V_O^{\min} - A_O$ (goal-feasibility diagnostic; positive means *unmet* under the best available policy), and the *control regret* $\delta_{\text{regret}} := A_O - V_O(M_t, \pi_{\text{current}}; N_h) \geq 0$ (policy-quality diagnostic). The 2×2 cross of these two gaps routes corrective action: *Success* (no action), *Strategy problem* (revise the policy), *Capability limit* (revise the model, policy class, or horizon — in that order), *Both* (revise the policy first, then re-evaluate). Without this routing, an agent receiving a regret signal cannot distinguish “the goal is unattainable from M_t ” from “the current policy is suboptimal.” A single $\delta_{\text{objective}} = V_O^{\min} - V_O(M_t, \pi_{\text{current}}; N_h)$ collapses both regimes into a single number and loses the diagnostic. The two-gap split runs along an axis *orthogonal* to the exploration-vs-adaptation decompositions of [5–7]: theirs isolates uncertainty *sources*; ours isolates *what is wrong*.

Component 2: Point-mass reverse-KL/TV identity — metric layer on policy space Under deterministic $\pi^* = \delta_{a^*}$, the reverse-KL and total variation between π^* and any policy Q are related by the *exact identity* $D_{\text{KL}}(\pi^* \parallel Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q)) \iff \text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \parallel Q)}$. This is a two-line direct calculation under the point-mass assumption (full statement and proof in Section 5.1). It is *not* the Bretagnolle–Huber inequality at equality: substituting it into the BH form $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ gives $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, *strictly* on $D_{\text{KL}} > 0$. The identity sits *strictly below* the BH envelope at this corner. Composed with the textbook TV-regret bound $R(Q) \leq V_{\max} \text{TV}(\pi^*, Q)$ and the matching action-gap lower bound, it yields the two-sided characterization $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi^* \parallel Q)}) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi^* \parallel Q)})$, *Lipschitz-equivalent* with constants $\Delta_{\min}/V_{\max}$ and 1. The identity coordinate is *coordinate-optimal among bounds depending only on TV*: any tighter bound on $R(Q)$ requires information beyond TV (the value-landscape spread). Strict improvement over Pinsker is uniform: at $D_{\text{KL}} = 0.01$ the identity bound is $7 \times$ tighter; for $D_{\text{KL}} > 2$ Pinsker is vacuous against the trivial $V_{\max}$ envelope while the identity saturates at $V_{\max}$. The deterministic- π^* scope is not a hard wall: the identity extends *perturbatively* to ϵ -greedy and softmax-regularized optima with corrections $O(\epsilon \log(1/\epsilon))$ and $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ respectively — exponentially small with polynomial prefactor in the softmax case (Section 5.1).

Component 3: Multi-factor strategic tempo with forgetting prerequisite — non-stationarity persistence Consider a structured policy with $|E|$ revisable elements (in our analysis, edges of an internal causal model the agent maintains over policy components; in Lee et al. 2023 [8]’s ProST, sub-policies indexed by a tempo schedule). For each element (i, j) , let ν_{ij} denote the per-element observation rate, $\iota_{ij} \in [0, 1]$ the identifiability coefficient (fraction of evidence that genuinely identifies the causal effect rather than reflecting confounded association), and $\eta_{\text{edge}, ij}$ the per-element update gain. Within a sector-Lyapunov reduction with diagonal correction (full Model (S) in Section 5.2), per-element exponential forgetting at rate λ_{ij} gives steady-state gain $\eta_{\text{edge}, ij}^{\text{ss}} \approx 1 - \lambda_{ij}$, and ultimate boundedness of the strategic mismatch $\|\delta_{\Sigma}\|$ within the modeled dynamics is *sufficient* under the bottleneck threshold $\mathcal{T}_{\Sigma}^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_{\Sigma}/R_{\Sigma}$. All three factors are multiplicatively load-bearing per element (zeroing any collapses the bottleneck; low identifiability cannot be compensated by faster forgetting), and the *bottleneck* (per-element minimum, not aggregate sum) is the right aggregator: a Lyapunov derivation under adversarial disturbance saturates at min,

not $\sum$, when the disturbance concentrates on the weakest element. The structural-class theorem identifies the *gain-decay* class $\mathcal{A}_{\text{decay}}$ — count-accumulating Bayesian updates without forgetting (e.g., Beta-Bernoulli with $\eta = 1/(n+1)$), bounded-memory schemes with growing memory, observation-aggregating schemes without restart, and gradient-based methods with vanishing step sizes (the Robbins–Monro regime, $\eta_t = c/t^\beta$ for $\beta \in (0, 1]$) — as one in which *every* agent eventually violates the threshold for any positive disturbance rate (formal statement in Section C.6). Non-accumulating mechanisms (constant-step SA, sliding-window, bounded-memory, block-restart) face *bidirectional* ceilings: the threshold is tight in both directions and the schedule has to be in the right regime, not arbitrarily fast or slow. The familiar single-factor form $(1 - \lambda) > \rho_\Sigma / R_\Sigma$ is recovered as a corollary under uniform-Regime-A Beta-Bernoulli normalization.

Component 4: Closed-loop interventional access — on-policy learnability Pearl’s causal hierarchy^[35] and the causal-hierarchy theorem of^[29] establish that Level-2 (interventional) queries $P(Y \mid \text{do}(X))$ are not in general identifiable from Level-1 (associational) data $P(Y \mid X)$. By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes o_{t+1} . Under three conditions on the agent’s action mechanism — (C1) *positivity* ($\pi_t(a \mid H_t) \geq p_{\min} > 0$ on the support of the information state H_t), (C2) *sequential ignorability* (the information state blocks unobserved confounding of action selection), and (C3) *known action mechanism* (the behavior policy is known) — conditioning on H_t blocks the action-selection confounding pathway, so the conditional distribution $P(o_{t+1} \mid a_t, H_t)$ coincides with the interventional distribution $P(o_{t+1} \mid \text{do}(a_t), H_t)$. The loop’s data-generating process is *interventional* by construction (formal statement in Section 5.3). This makes the regret bound of Component 2 *learnable on-policy* in regimes with sufficient identifiability: the agent’s identified optimum $a_{\text{ag},t}^*$ matches the true optimum $\tilde{a}_t^*$ with probability p_{id} , and the per-state, per-step bias from misidentification is bounded by $(1 - p_{\text{id}}(s)) \log(1/q_0)$ under two-point support $Q_t \geq q_0$ (Section 5.4). Three regimes^[29] partition usable strength: A (intervention-rich, $\iota \approx 1$), B (partial, $\iota \in (0, 1)$), C (observation-only, $\iota \approx 0$).

4.2 The composition theorem

Theorem 4.1 (Composition: joint guarantees with conditional cumulative reduction). *Let $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$ be a non-stationary MDP with bounded reward, finite action space, and for each round t a deterministic optimum policy $\pi_t^* = \delta_{a_t^*}$ with action gap $\Delta_{\min} > 0$. Suppose the agent operates on a singular causal trajectory in the sense of Section 3 and updates its policy with per-element exponential forgetting at rates $\{\lambda_{ij}\}$. Let $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$ be the per-state reverse-KL coordinate (in the bandit case $N_h = 1$ the state argument is suppressed). If*

(A1) **Metric on policy space.** $Q_t(a_t^* \mid s) > 0$ at every visited state and round.

(A2) **Multi-factor forgetting prerequisite.** Within Model (S) of Section 5.2, the bottleneck strategic tempo $\tau_{\Sigma}^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$.

(A3) **Identifiability regime.** The action mechanism is known and (C1)–(C3) of Section 5.3 hold; the agent operates in Regime A or Regime B with per-state argmax-correctness probability $p_{\text{id}}(s)$, with floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$.

(A4) **Two-gap diagnostic.** The agent applies the satisfaction-gap / control-regret 2×2 to route corrective action.

(A5) **Piecewise-stationary block structure with restart-on-change identity-tight base learner.** The time axis partitions into $B_T + 1$ stationary intervals on which the MDP is stationary and $\pi_t^*(\cdot \mid s) = \delta_{a_t^*(s)}$ is fixed per state. The base learner is restarted at each block boundary; within each restarted interval of length L it satisfies $\sum_{t=1}^L \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{L}$, where $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid s_h), Q_t(\cdot \mid s_h))]$ is the occupancy-weighted per-round coordinate. (A non-restarting carryover variant is plausible but requires bounding cross-block occupancy shift; see Remark in Section 5.)

Then:

(i) **Per-round regret is two-sided identity-bounded.** At every visited state, $\Delta_{\min}(1 - e^{-K_t(s)}) \leq R(Q_t \mid s) \leq V_{\max}(1 - e^{-K_t(s)})$.

334 **(ii) Aggregate strategic mismatch is ultimately bounded.** Under (A2), the modeled mismatch $\|\delta_\Sigma\|$ is
335 ultimately bounded within $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$.

336 **(iii) The KL coordinate is computable with controlled bias.** $-\log Q_t(a_{\text{ag},t}^*(s) \mid s)$ is computable
337 directly from Q_t . Per-state bias: $\mathbb{E}[\|\hat{D}_t(s) - D_t^{\text{true}}(s)\|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$ under $Q_t(\cdot \mid s) \geq q_0$
338 at the two argmax candidates. Vanishes in Regime A.

339 **(iv) The 2×2 cell identifies the corrective-action class.** $(\delta_{\text{sat}}, \delta_{\text{regret}})$ routes to revise-policy / revise-
340 model-class-or-horizon / revise-objective.

341 **(v) Trajectory-level blockwise dynamic regret.** Under (A5), $\mathbb{E}[\text{DynReg}(T)] \quad :=$
342 $\mathbb{E}\left[\sum_{t=1}^T (V_t^{\pi_t^*} - V_t^{Q_t})\right] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + N_h(1 - p_{\text{id}}) \log(1/q_0) \cdot T.$

343 The proof composes the per-round identity (i) with the simulation lemma^[40–43], a block decomposition
344 at optimum-change events, Cauchy–Schwarz across the $B_T + 1$ blocks, and the bias bound from (iii).
345 Full proof in Section A; the four key lemmas are surfaced in Section 5.

346 4.3 Unpacking the conclusions

347 Theorem 4.1 asserts that the four components compose into a non-stationarity-aware convergence
348 theory with three properties that no existing strand has individually: it *handles non-stationarity*
349 (via (ii)/(A2)), has *explicit metric structure on policy space* (via (i)/Component 2), and is *learnable*
350 *on-policy* (via (iii)/(A3)). Conclusion (iv) is the connective tissue routing corrective action; conclusion
351 (v) is the cumulative regret rate that follows from composing the per-round metric coordinate with the
352 base-learner stationary rate.

353 We emphasize four things about this theorem.

354 *The per-round coordinate is exact, not an inequality.* Under the canonical scope (deterministic π^*), the
355 per-round identity-bound (i) is $\Delta_{\min}/V_{\max}$ -Lipschitz-equivalent to the regret. Both endpoints of the
356 Lipschitz envelope are *achieved* on extremal value landscapes ($\Delta(a) = V_{\max}$ for all $a \neq a^*$ saturates
357 the upper bound; $\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$ saturates the lower). The per-round coordinate is therefore
358 *coordinate-optimal among bounds depending only on TV*: tightening would require value-landscape
359 information beyond TV. This is sharper than Pinsker’s $\sqrt{D_{\text{KL}}/2}$ — exact instead of approximate —
360 and sharper than the Bretagnolle–Huber envelope $\sqrt{1 - e^{-D_{\text{KL}}}}$ at this corner since $x < \sqrt{x}$ on $(0, 1)$.

361 *The cumulative rate is in the piecewise-stationary B_T family, not the continuous-variation V_T family.*
362 The lifted rate $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ that follows under UCRL2/UCBVI base learners (Section D)
363 sits in the same block-shape family as the restart-on-change variants of^[1] — the route here is through
364 the per-round identity rather than direct optimism. It is *not directly comparable* to the continuous-
365 variation rate $\tilde{O}(SA V_T^{1/3} H T^{2/3})$ of^[31]: the parameter (B_T vs. V_T), the T -scaling ($T^{1/2}$ vs. $T^{2/3}$),
366 the S, A -scaling ($\sqrt{SA}$ vs. $(SA)^{1/3}$), and the horizon dependence (N_h^2 vs. N_h) all differ. Under
367 abrupt-magnitude- δ piecewise-stationarity the two regimes coincide up to constants ($V_T \asymp \delta B_T$),
368 and B-CS1 is sharper for fixed B_T ; outside that, Mao-style continuous-variation analysis is sharper
369 for fixed V_T . Generalizing the per-round-identity route to the continuous-variation regime is open
370 (Section 6).

371 *The bias term in (v) is well-controlled in Regimes A/B and saturates in Regime C.* The second term
372 $N_h(1 - p_{\text{id}}) \log(1/q_0) \cdot T$ is linear in T — vacuous as a rate. It vanishes in Regime A ($p_{\text{id}} \rightarrow 1$ under
373 (C1)–(C3)). In Regime B ($p_{\text{id}} \in (0, 1)$), the bias is controlled but linear; the first term ($\sqrt{(B_T + 1)T}$)
374 is sublinear, so the dominant rate term is the metric one provided $1 - p_{\text{id}}$ is small enough. In Regime
375 C ($p_{\text{id}} \rightarrow 0$), the bound is provable in principle but vacuous as an actionable rate; algorithm flags this
376 regime as out-of-scope for on-policy learnability (Section D).

377 *The horizon factor is the standard simulation-lemma penalty.* The N_h in conclusion (v) is the linear-
378 in- N_h per-round simulation-lemma coefficient — sharper than the N_h^2 of TRPO-style worst-state
379 bounds^[44]. The lifted N_h^2 in the UCRL2/UCBVI cumulative rate reflects the additional N_h in the
380 within-block base-learner constant c .

381 4.4 Necessity of the four components

382 A natural reviewer question: why exactly four? The composition is a *bundle of compatible guarantees*
 383 (each conclusion supported by its own assumptions), not a single integrated theorem in which every
 384 component is necessary for the rate. The rate (v) goes through (A5) and (i) alone; the joint properties
 385 — rate plus persistence plus learnability plus corrective-action routing — require all four.

386 **Without Component 2 (point-mass identity):** the per-round bound is Pinsker $V_{\max} \sqrt{D_{\text{KL}}/2}$ or BH
 387 $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$ — both strictly looser at this corner; Pinsker is vacuous for $D_{\text{KL}} > 2$. The metric
 388 layer loses exactness, and the “behavior-cloning loss against optimal trajectory” interpretation of the
 389 cumulative coordinate vanishes.

390 **Without Component 3 (strategic tempo):** the agent has no persistence guarantee. By the structural-
 391 class theorem on $\mathcal{A}_{\text{decay}}$, every count-accumulating update without forgetting eventually violates
 392 the threshold for any positive disturbance; the modeled mismatch $\|\delta_{\Sigma}\|$ grows unboundedly. The
 393 per-round bound (i) still holds, but $K_t(s)$ may itself drift — there is no guarantee the per-round
 394 coordinate stays bounded.

395 **Without Component 4 (loop-Level-2):** the bound’s RHS is computable from Q_t , but its interpreta-
 396 tion as regret-against-true-optimum requires identifying $a_t^*(s)$ — an interventional query under the
 397 environment’s transition kernel. In Regime C ($\iota \approx 0$), the bias saturates at $\log(1/q_0)$ per state per
 398 step, contributing $N_h \log(1/q_0) \cdot T$ to cumulative regret — the rate (v) becomes vacuous.

399 **Without Component 1 (two-gap diagnostic):** the regret signal δ_{regret} alone does not distinguish
 400 “policy revision is the right move” from “the goal is unattainable from M_t .” An agent might persist in
 401 policy revision when the model, policy class, or horizon is the actual problem (Capability-limit cell),
 402 or revise the goal when policy revision would have sufficed.

403 So while the rate (v) goes through (A5) and (i) alone, the *bundle* — rate plus persistence plus
 404 learnability plus corrective-action routing — requires all four. Each drop is a concrete loss of an
 405 epistemic property that the field treats as separate.

406 5 Mechanism

407 This section overviews the proof of Theorem 4.1 via four key lemmas chained through a block
 408 decomposition. Each lemma is stated formally; intuitions are given in plain English with the algebra
 409 deferred to the appendices. The rate (v) is the composition’s load-bearing conclusion; we sketch its
 410 derivation in four steps after surfacing the lemmas.

411 5.1 Key Lemma 1: Point-mass reverse-KL/TV identity

412 **Lemma 5.1** (Point-mass reverse-KL/TV identity). *For deterministic $\pi^* = \delta_{a^*}$ and any policy*
 413 *Q with $Q(a^*) > 0$, $D_{\text{KL}}(\pi^* \| Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q))$, equivalently*
 414 *$\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$.*

415 *Intuition.* The reverse-KL collapses under the point-mass: $D_{\text{KL}}(\delta_{a^*} \| Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)}$,
 416 where the convention $0 \log 0 = 0$ kills every $a \neq a^*$ term, leaving $\log \frac{1}{Q(a^*)} = -\log Q(a^*)$. Total
 417 variation collapses just as cleanly: $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2} \sum_a |\delta_{a^*}(a) - Q(a)| = 1 - Q(a^*)$. Combining
 418 gives the identity. Substituting into the Bretagnolle–Huber inequality $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$
 419 yields $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, strictly on $D_{\text{KL}} > 0$. The identity is therefore not BH-at-equality;
 420 it is the *exact value* at the deterministic- π^* corner, sitting strictly below the BH envelope. Full proof
 421 in Section B.

422 *Per-round regret bound.* Composed with the textbook TV-regret bound $R(Q) \leq V_{\max} \text{TV}(\pi^*, Q)$
 423 and the matching action-gap lower bound $R(Q) \geq \Delta_{\min} \text{TV}(\pi^*, Q)$, the identity yields the two-sided
 424 characterization that gives Theorem 4.1’s conclusion (i). The bound is *coordinate-optimal among*
 425 *bounds depending only on TV*: any tighter bound on $R(Q)$ requires value-landscape information
 426 beyond TV.

427 *Perturbative extension.* The deterministic- π^* scope is not a hard wall. For ϵ -greedy π_ϵ^* with Q
428 satisfying full-support $Q(a) \geq q_0$, $\text{TV}(\pi_\epsilon^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)} + O(\epsilon \log(1/\epsilon))$ uniformly in Q ;
429 the proof is a triangle-inequality + expansion argument given in Section B.2. For softmax-regularized
430 $\pi_\tau^* \propto \exp(Q_O/\tau)$, the off-optimum mass concentrates as $O(\exp(-\Delta_{\min}/\tau))$ and the correction is
431 $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ — exponentially small with polynomial prefactor. The two-sided regret bound
432 transfers with the same correction order. Outside the perturbative regime — genuinely high-entropy
433 optima, tied-optimum, hard-exploration — the BH inequality becomes the relevant general bound;
434 Pinsker is the textbook fallback.

435 5.2 Key Lemma 2: Forgetting prerequisite

436 The strategic mismatch state $\delta_\Sigma = (\delta_{ij})_{(i,j) \in E}$ in the Euclidean norm $\|\delta_\Sigma\|^2 = \sum_{(i,j)} \delta_{ij}^2$ evolves
437 in discrete time as $\mathbb{E}[\Delta \delta_{ij} \mid \delta_{ij}] = -\alpha_{ij} \delta_{ij} + w_{ij}$, $|w_{ij}| \leq \rho_\Sigma/|E|^{1/2}$, where $\alpha_{ij} \geq 0$ is
438 a per-element correction rate (the strategic sector parameter), w_{ij} is a bounded disturbance, and
439 the disturbance budget satisfies $\sum_{(i,j)} w_{ij}^2 \leq \rho_\Sigma^2$. The strategic reserve R_Σ is the operative norm-
440 radius beyond which the agent loses tracking. Beta-Bernoulli edge updates instantiate this *Model (S)*
441 approximately; gradient-based decaying-step updates instantiate it under standard Robbins–Monro
442 conditions; we will refer to it as *Model (S)* throughout.

443 **Lemma 5.2** (Multi-factor forgetting prerequisite). *Within Model (S) with diagonal per-element*
444 *correction rates $\alpha_{ij} = \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij}$ (per-element observation rate $\nu_{ij} \in [0, 1]$, regime-adjusted*
445 *identifiability $\iota_{ij} \in [0, 1]$, edge-update gain $\eta_{\text{edge},ij} > 0$), under per-element exponential forgetting*
446 *at rate $\lambda_{ij} \in (0, 1)$ giving steady-state $\eta_{\text{edge},ij}^{\text{ss}} \approx 1 - \lambda_{ij}$, ultimate boundedness of $\|\delta_\Sigma\|$ within*
447 *$R_\Sigma^* = \rho_\Sigma/\mathcal{T}_\Sigma^{\text{bn,ss}}$ is sufficient under $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} (\nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})) > \rho_\Sigma/R_\Sigma$. The*
448 *threshold is sharp inside the diagonal sector model: when the inequality reverses, an adversarial*
449 *disturbance concentrating on the bottleneck element drives $\|\delta_\Sigma\|$ beyond R_Σ within the modeled*
450 *dynamics.*

451 *Intuition (why bottleneck, not sum).* The Lyapunov sector product is $\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) = \sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2$.
452 Bounding this from below uniformly in δ_Σ requires the largest constant c such that $\sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2 \geq$
453 $c \cdot \|\delta_\Sigma\|^2$ holds for all δ_Σ . The largest such c is $\min_{(i,j)} \alpha_{ij}$, not $\sum_{(i,j)} \alpha_{ij}$ — adversarial concentration
454 of the disturbance on the *weakest* element saturates the bound at the bottleneck. The same effect
455 makes per-dimension persistence conditions sharper than scalar conditions in adaptive control^[45]:
456 scalar tempo can overstate the available margin because the worst-case disturbance concentrates
457 on the slowest-correcting dimension. Ultimate boundedness then follows from Khalil’s classical
458 sector-Lyapunov template applied to the strategic substate. Full proof in Section B; the structural-class
459 theorem on $\mathcal{A}_{\text{decay}}$ (gain-decay updates universally violate the threshold) is in Section C.6.

460 *ProST as a sector-level reduction.* The block-restart structure of ProST^[8] idealizes as an *impulsive*
461 *system* — between scheduled update times the policy is held and the mismatch evolves under dis-
462 turbance alone (continuous-destabilizing in the Lyapunov $V = \|\delta_\Sigma\|^2$); at each update time the
463 policy is revised, contracting the modeled mismatch by per-update impulse gain $\gamma \in (0, 1]$. The
464 reverse-average-dwell-time framework of^[46] (Theorem 1, reverse-ADT branch) gives the threshold
465 $\Delta_{\max} \cdot \rho_\Sigma/R_\Sigma < -\ln(1 - \gamma)^2$ where $\Delta_{\max}$ is the longest inter-update gap. Under uniform blocks
466 $\Delta_i = T/K$ this reduces to $K/T > \rho_\Sigma/(2\gamma R_\Sigma)$ to leading order in γ — recovering ProST’s K/T
467 form with the impulse gain made explicit. Under nonuniform schedules the impulsive form is *strictly*
468 *stronger*: the longest block sets the threshold, not the average. This makes a tradeoff between update
469 frequency $1/\Delta_{\max}$ and per-update impulse strength γ visible at the sector-Lyapunov level — the same
470 disturbance can be tolerated either by frequent weak updates or rare strong ones.

471 5.3 Key Lemma 3: Closed-loop interventional access

472 **Lemma 5.3** (Loop generates interventional samples). *Let H_t denote the agent’s information state*
473 *at round t (model state M_t , history of observations and actions). Under (C1) positivity — $\pi_t(a \mid$
474 $H_t) \geq p_{\min} > 0$ on the support of H_t for the actions of interest; (C2) sequential ignorability —
475 H_t blocks unobserved confounding of the action mechanism, so $a_t \perp\!\!\!\perp (\text{environment latents}) \mid H_t$;
476 (C3) known action mechanism — the behavior policy $\pi_t(a \mid H_t)$ is known: executed actions are
477 interventions relative to the environment’s transition kernel, and the loop generates samples from*

478 $P(o_{t+1} \mid \text{do}(a_t = a), H_t)$ — i.e., *Pearl Level-2 interventional kernels conditional on the decision*
 479 *context.*

480 *Intuition.* By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally
 481 precedes o_{t+1} . Under (C1)–(C3), conditioning on the information state H_t blocks the action-selection
 482 confounding pathway, so the conditional distribution $P(o_{t+1} \mid a_t, H_t)$ coincides with the interven-
 483 tional distribution $P(o_{t+1} \mid \text{do}(a_t), H_t)$ on the support where $\pi_t(a_t \mid H_t) > 0$. The architecture
 484 of the agent (Q-learning, transformer, etc.) does not enter: the loop generates conditional-Level-2
 485 samples whenever (C1)–(C3) hold, irrespective of how the agent updates internally. Whether the
 486 agent *exploits* this data for Level-2 reasoning is a separate algorithmic question.

487 *Three regimes of identifiability.* Three regimes^[29] partition usable strength of identification: - **Regime**
 488 **A (intervention-rich, $\iota \approx 1$).** Software tests, controlled labs. do-effects identified cleanly; bound
 489 realizable on-policy. - **Regime B (partial, $\iota \in (0, 1)$).** Mixed observation-intervention. Bound
 490 holds for the model the agent identifies; bias $\propto 1 - \iota$. - **Regime C (observation-only, $\iota \approx 0$).**
 491 Passive monitoring. Bound provable analytically but not realizable on-policy; observer-on-sub-agent
 492 extensions can recover identifiability in subcases (Section C.8).

493 Action-generated data is Level 2 in *character*, but a clean estimate of $P(o \mid \text{do}(a), H_t)$ requires
 494 overcoming four typical obstacles: coverage (diverse actions tried), within-step confounding, delay
 495 (consequences past $t + 1$), and partial observability. We honor this distinction. Components 2 and
 496 4 are jointly load-bearing: the identity of Section 5.1 gives the analytic regret-KL relationship; the
 497 loop-Level-2 access here gives the data substrate from which π^* — and the bound’s RHS — can be
 498 empirically estimated under sufficient identifiability. Neither alone makes the bound learnable.

499 5.4 Key Lemma 4: Bias bound under partial identifiability

500 **Lemma 5.4** (Bias bound). Assume $Q_t(a \mid s) \geq q_0 > 0$ for both $a = a_{\text{ag},t}^*(s)$ (the agent’s identified
 501 optimum) and $a = \tilde{a}_t^*(s)$ (the true optimum). Let $\hat{D}_t(s) := -\log Q_t(a_{\text{ag},t}^*(s) \mid s)$ and $D_t^{\text{true}}(s) :=$
 502 $-\log Q_t(\tilde{a}_t^*(s) \mid s)$. Then $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$, and taking expecta-
 503 tions: $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$, where $p_{\text{id}}(s) := \Pr[a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)]$.

504 *Intuition.* The argument is an indicator decomposition. On the matching event, the difference is
 505 identically zero. On the misidentification event, $\log Q_t(a_{\text{ag},t}^*(s) \mid s)$ and $\log Q_t(\tilde{a}_t^*(s) \mid s)$ both lie
 506 in $[\log q_0, 0]$, so their absolute difference is at most $\log(1/q_0)$. The expectation bound follows from
 507 $\Pr[\text{misidentification event}] = 1 - p_{\text{id}}(s)$. Bias scales *linearly* in misidentification mass — vanishes
 508 in Regime A ($p_{\text{id}} \rightarrow 1$), controlled in Regime B, saturates at $\log(1/q_0)$ in Regime C. The argmax-
 509 correctness probability p_{id} relates to the per-edge identifiability ι via the policy DAG structure; in
 510 the simple-bandit case $p_{\text{id}} = \iota$. Note: the support condition is *two-point* (only at the two argmax
 511 candidates), strictly weaker than the *full-support* condition $Q(a) \geq q_0$ for all a that the perturbative
 512 identity (Section 5.1) requires — F.1 applies whenever the two candidates are positively-supported
 513 regardless of Q_t ’s mass on other actions. Full proof in Section B.

514 5.5 Combining the lemmas: cumulative regret in four steps

515 We sketch the proof of conclusion (v) of Theorem 4.1 — the cumulative trajectory-level dynamic
 516 regret rate — by chaining the four key lemmas through a block decomposition.

517 *Step 1 — Per-round bound via the simulation lemma.* For finite-horizon N_h non-stationary MDPs with
 518 bounded reward, the standard performance-difference / simulation lemma^[40–43] gives, for any two
 519 policies π_t^* and Q_t and initial state s_0 , $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid$
 520 $s_h), Q_t(\cdot \mid s_h))]$ = $V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$, where $d_h^{Q_t}$ is the round- t learner-induced state distribution and
 521 $\overline{\text{TV}}_t$ is the occupancy-weighted TV defined in (A5). By Key Lemma 1 applied per state, $\text{TV}(\pi_t^*(\cdot \mid$
 522 $s), Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$ — strictly sharper than Pinsker — preserved underneath the occupancy
 523 aggregation. The horizon factor N_h here is the linear-in- N_h simulation-lemma penalty, sharper than
 524 the N_h^2 of TRPO-style worst-state bounds^[44].

525 *Step 2 — Block decomposition at optimum-change events.* Partition $[1, T]$ at optimum-change events
 526 into $0 = \tau_0 < \tau_1 < \dots < \tau_{B_T} \leq \tau_{B_T+1} = T$. Within each block $[\tau_i, \tau_{i+1})$ of length Δ_i

the MDP is stationary. (A5) — restart-on-change base learner — gives within-block guarantee $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$.

Step 3 — Cauchy–Schwarz across blocks. Across the $B_T + 1$ blocks, $\sum_i \sqrt{\Delta_i} \leq \sqrt{(B_T + 1)T}$ by Cauchy–Schwarz. Multiplying by $V_{\max}N_h$ closes the rate term: $\sum_t \mathbb{E}[V_{\max}N_h \overline{\text{TV}}_t] \leq 2cV_{\max}N_h \sqrt{(B_T + 1)T}$.

Step 4 — Bias term via Key Lemma 4. The bias from misidentification at each visited state per horizon step contributes at most $(1 - p_{\text{id}}(s)) \log(1/q_0)$ per state-step (Key Lemma 4, per-state form). Aggregated linearly over the N_h horizon steps and the T rounds with the per-state floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$, this contributes $N_h(1 - p_{\text{id}}) \log(1/q_0) \cdot T$ — vanishes in Regime A.

Combining Steps 1–4 gives conclusion (v): $\mathbb{E}[\text{DynReg}(T)] \leq 2cV_{\max}N_h \sqrt{(B_T + 1)T} + N_h(1 - p_{\text{id}}) \log(1/q_0) \cdot T$. The Cesàro tracking corollary $\frac{1}{T} \sum_t (V_t^* - V(\pi_t)) = O(N_h \sqrt{(B_T + 1)/T}) \rightarrow 0$ (when $B_T = o(T)$ and $p_{\text{id}} \rightarrow 1$) follows by dividing through by T .

Bandit case sharpening In the bandit special case ($N_h = 1$), Thompson sampling and UCB satisfy (A5) with the *stronger logarithmic rate* $\mathbb{E}[1 - e^{-K_t}] = O(\log t / (t \Delta_{\min}))$ per stationary block (Section D), giving cumulative regret $O(V_{\max}(B_T + 1) \log^2(T/(B_T + 1))/\Delta_{\min})$ — sharper than $\sqrt{(B_T + 1)T}$ when $\Delta_{\min}$ is bounded away from zero. The square-root rate is the worst-case Cauchy–Schwarz bound; the logarithmic rate is the typical-case bound for stochastic-bandit base learners.

Carryover variant of (A5) If the base learner does *not* restart at block boundaries — e.g., a single UCRL2 instance run across the whole horizon — the within-block $\sqrt{L}$ guarantee can fail at switch events because $d_h^{Q_t}$ at the start of block $i + 1$ reflects the previous block’s policy rather than a fresh exploration distribution. Bounding the cross-block occupancy shift requires either (a) detecting switches and locally re-exploring (which recovers a restart-style analysis), or (b) showing the base learner’s regret is robust to mis-calibrated state distributions. Sliding-window or adaptive-window variants^[1,2] can be analyzed this way under additional structural assumptions. Theorem 4.1(v) is stated for the restart case; the carryover case is plausible but requires bounding the occupancy shift, deferred as future work.

Detailed full proof of (v) in Section A.

6 Conclusion

We assemble four components — the two-gap diagnostic, the point-mass reverse-KL/TV identity, the multi-factor strategic-tempo forgetting prerequisite, and closed-loop interventional access — into a unified non-stationary RL convergence theory. The composition delivers three properties no published strand has individually: it *handles non-stationarity* (via the strategic-tempo persistence inequality), has *explicit metric structure on policy space* (via the point-mass identity), and is *learnable on-policy* (via closed-loop interventional access). The cumulative dynamic regret rate $\tilde{O}(N_h \sqrt{(B_T + 1)T})$ matches the piecewise-stationary B_T family of^[1] as a corollary; the per-round identity coordinate is *exact* under deterministic π^* and strictly sharper than Pinsker and Bretagnolle–Huber at this corner. The point-mass identity is the technical anchor — absent, to our knowledge, from the prior RL/non-stationary corpus as an upper-bound regret coordinate.

We close with concluding observations and practitioner takeaways.

6.1 Concluding observations

On the optimal dependencies on N_h , S , A The lifted UCRL2/UCBVI rate $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ has linear-in- N_h per-round penalty (the simulation lemma) plus an additional N_h from the within-block constant c . Bernstein-type concentration in the base learner could potentially shave one $\sqrt{N_h}$ factor, by analogy with^[43] for tabular non-stationary MDPs. We believe the $\sqrt{N_h}$ gap between this and the corresponding tabular non-stationary lower bound is expected because the base-learner concentration used here is intrinsically Hoeffding-type.

574 **On the continuous-variation extension** The current rate is in the piecewise-stationary B_T fam-
 575 ily. Lifting to the continuous-variation V_T family of^[3] would require a different block structure —
 576 variation-aware windows with adaptive width^[2] — and a different analysis at the per-round level (con-
 577 tinuous variation cannot be partitioned at discrete optimum-change events). It remains an interesting
 578 open question whether the per-round-identity route can match the $T^{2/3}$ continuous-variation rate, or
 579 whether the regimes are structurally distinct.

580 **On joint uniqueness of the four-component composition** The metric layer is uniquely reverse-
 581 KL up to positive scaling under chain-rule additivity (Theorem C.3 in Section C.10) — any theory
 582 satisfying explicit metric structure plus chain-rule must use reverse-KL. We do *not* claim joint
 583 uniqueness across all three properties (non-stationarity / metric / learnable). Alternative correction
 584 architectures (non-sector-bounded revisions; non-diagonal correction) and alternative interventional-
 585 access taxonomies exist in principle, and a joint-uniqueness theorem would require further axioms
 586 (singular-trajectory + sector-boundedness + identifiability conditions) — open.

587 **On coupled-goal architectures** (C2)’s sequential-ignorability assumption presupposes architec-
 588 tural separation (conditional independence) between the model state M_t and the goal state. Goal-
 589 conditioned LLM policies and similar coupled-goal architectures need additional machinery to satisfy
 590 (C2); the conditional-independence relaxation results of^[47] are a starting point. We treat this as an
 591 interesting boundary case that the current framework does not directly cover.

592 **On strict tightness of the strategic-tempo threshold** The bottleneck threshold $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$ is
 593 *sufficient and sharp inside the diagonal sector model* — when reversed, an adversarial disturbance
 594 concentrating on the bottleneck element drives the modeled mismatch beyond R_{Σ} . The theorem
 595 is silent about correction architectures *outside* the sector-Lyapunov reduction (e.g., non-diagonal
 596 correction, alternative non-sector-bounded revisions, or stabilization produced by mechanisms other
 597 than the modeled α_{Σ} -correction). Whether the threshold extends to those architectures is a useful
 598 open question in the adaptive-control-RL bridge.

599 6.2 Practitioner takeaways

600 For a practitioner seeking to apply the four-component theory:

- 601 1. **The point-mass identity at the deterministic- π^* corner is a corner-case sharpening**
 602 **of Bretagnolle–Huber that hasn’t been deployed in RL regret upper bounds.** Where
 603 deterministic π^* applies — most tabular finite-MDP regret analyses with isolated optima —
 604 the identity yields *exact* regret-vs-KL Lipschitz equivalence rather than the loose Pinsker /
 605 BH inequalities. The per-round coordinate $1 - e^{-D_{\text{KL}}}$ is computable directly from Q and is
 606 strictly sharper than $\sqrt{D_{\text{KL}}/2}$ at every $D_{\text{KL}} > 0$.
- 607 2. **The multi-factor strategic tempo lifts ProST’s tempo-as-hyperparameter framing into**
 608 **a structural survival inequality.** Three factors (observation rate ν , identifiability ι , discount
 609 rate $1 - \lambda$) are independently load-bearing per element. A practitioner can use the aggregate
 610 necessary-condition $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$ as a fail-fast pre-check and the variation-budget
 611 estimation literature^[1,2,39] to estimate the environment-side ρ_{Σ}/R_{Σ} from observed reward
 612 and transition variation.
- 613 3. **Closed-loop interventional access makes regret bounds learnable on-policy under suffi-**
 614 **cient identifiability.** Regime A (intervention-rich, $\iota \approx 1$) makes the bound directly realizable;
 615 Regime B (partial) carries quantified bias linear in $1 - p_{\text{id}}$; Regime C (observation-only)
 616 requires observer-on-sub-agent extensions (Section C.8) or flags the regime as out-of-scope.
 617 The 2×2 diagnostic routes corrective action when regret is non-zero — distinguishing “policy
 618 revision” from “model / policy class / horizon revision” from “objective revision.”

619 We hope this work contributes a unifying lens for the four-track non-stationary RL landscape, and
 620 that the point-mass identity is a useful tool in subsequent regret analyses where deterministic π^* is
 621 the canonical scope.

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A Proof of the Composition Theorem

This appendix gives the full proof of Theorem 4.1. The proof composes the four key lemmas of Section 5 (full proofs of those in Section B) with a block decomposition at optimum-change events plus Cauchy–Schwarz across blocks.

Notation Throughout this section: $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$ is the per-state reverse-KL coordinate; $d_h^{Q_t}(\cdot \mid s_0)$ is the round- t learner-induced state distribution at horizon step h starting from s_0 ; $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid s_h), Q_t(\cdot \mid s_h))]$ is the occupancy-weighted per-round coordinate of (A5); $B_T = |\{t : a_t^* \neq a_{t-1}^*\}|$ counts optimum-change events; the partition is $0 = \tau_0 < \tau_1 < \dots < \tau_{B_T} \leq \tau_{B_T+1} = T$ with block i length $\Delta_i := \tau_{i+1} - \tau_i$.

A.1 Proof of (i) — Per-round identity-bounded regret

For deterministic $\pi_t^* = \delta_{a_t^*(s)}$ and any $Q_t(\cdot \mid s)$ with $Q_t(a_t^*(s) \mid s) > 0$ at every visited state (which is (A1)), Key Lemma 1 (Lemma 5.1, full proof in Section B.1) gives the exact identity $\text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$. Composed with the per-state TV-regret bounds (full proofs in Section B.1 and Section C.4): $\Delta_{\min} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) \leq R(Q_t \mid s) \leq V_{\max} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s))$, substituting the identity yields conclusion (i): $\Delta_{\min}(1 - e^{-K_t(s)}) \leq R(Q_t \mid s) \leq V_{\max}(1 - e^{-K_t(s)})$. $\square$

A.2 Proof of (ii) — Aggregate mismatch ultimate boundedness

Under (A2), Key Lemma 2 (Lemma 5.2, full proof in Section B.3) applied to the diagonal sector model with $\alpha_{ij}^{\text{ss}} = \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$ gives ultimate boundedness of $\|\delta_\Sigma\|$ within $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$. $\square$

A.3 Proof of (iii) — KL coordinate bias bound

Key Lemma 4 (Lemma 5.4, full proof in Section B.6) gives the per-state bias bound directly, under (A3)’s identifiability assumption (which gives $p_{\text{id}}(s)$ for each s) and the two-point support condition $Q_t(\cdot \mid s) \geq q_0$ at $a_{\text{ag},t}^*(s)$ and $\tilde{a}_t^*(s)$. $\square$

752 A.4 Proof of (iv) — 2×2 corrective-action routing

753 (A4) states the agent applies the satisfaction-gap / control-regret 2×2 (Section 4.1 Component 1).
 754 Conclusion (iv) is a direct restatement of the 2×2 cell partition. The decomposition is convention-
 755 dependent (the values of δ_{sat} and δ_{regret} depend on the continuation convention; see Section C.1 for
 756 monotonicity across C1/C2/C3); the 2×2 structure is preserved across all three conventions. $\square$

757 A.5 Proof of (v) — Trajectory-level cumulative regret

758 We chain Key Lemmas 1 and 4 with the simulation lemma and a Cauchy–Schwarz block argument.
 759 The proof is in four steps; the high-level sketch is in Section 5.5.

760 *Step 1: Simulation / performance-difference lemma.* For any two policies π
 761 and π' , any state s , and finite-horizon non-discounted MDP with horizon N_h ,
 762 the standard performance-difference lemma^[40–43] gives $V_t^{\pi'}(s) - V_t^\pi(s) =$
 763 $\mathbb{E}\left[\sum_{h=0}^{N_h-1} \left(\mathbb{E}_{a' \sim \pi'(\cdot | s_h)}[Q_h^{\pi'}(s_h, a')] - Q_h^\pi(s_h, a_h)\right) \mid s_0 = s, a_h \sim \pi\right]$. Each per-step bracket
 764 is bounded by value range times TV: $|\mathbb{E}_{a' \sim \pi'}[Q_h^{\pi'}] - \mathbb{E}_{a \sim \pi}[Q_h^\pi]| \leq V_{\max} \cdot \text{TV}(\pi'(\cdot | s_h), \pi(\cdot | s_h))$. Applied at round t with $\pi' = \pi_t^*$, $\pi = Q_t$: $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq$
 766 $V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}(\cdot | s_0)}[\text{TV}(\pi_t^*(\cdot | s_h), Q_t(\cdot | s_h))] = V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$. The horizon factor N_h
 767 here is the *linear-in- N_h* simulation-lemma penalty — sharper than the N_h^2 of TRPO-style worst-state
 768 bounds^[44], which bound max-state TV rather than occupancy-weighted TV.

769 *Step 2: Per-state TV identity along the trajectory.* By the deterministic-per-state optimum scope of
 770 Theorem 4.1 ($\pi_t^*(\cdot | s) = \delta_{a_t^*(s)}$), at every visited state Key Lemma 1 gives $\text{TV}(\pi_t^*(\cdot | s), Q_t(\cdot | s)) = 1 - Q_t(a_t^*(s) | s) = 1 - e^{-K_t(s)}$. Summing over the trajectory and taking expectations:
 772 $\overline{\text{TV}}_t = \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}}[1 - e^{-K_t(s_h)}]$. The per-state KL/TV identity is preserved underneath
 773 the occupancy aggregation; the per-round coordinate is sharper than Pinsker/BH at every visited state.

774 *Step 3: Block-sum + Cauchy–Schwarz across blocks.* Within each block $[\tau_i, \tau_{i+1})$ of length
 775 Δ_i , (A5)’s restart-on-change identity-tight base learner gives $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$. Sum-
 776 ming over the $B_T + 1$ blocks: $\sum_{t=1}^T \mathbb{E}[\overline{\text{TV}}_t] = \sum_{i=0}^{B_T} \sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c \sum_{i=0}^{B_T} \sqrt{\Delta_i}$.
 777 Cauchy–Schwarz gives $\sum_{i=0}^{B_T} \sqrt{\Delta_i} \leq \sqrt{(B_T + 1) \cdot \sum_i \Delta_i} = \sqrt{(B_T + 1)T}$. Multiplying by
 778 $V_{\max} N_h$: $\sum_{t=1}^T \mathbb{E}[V_{\max} N_h \overline{\text{TV}}_t] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T}$. This is the rate term.

779 *Step 4: Bias term via per-state, per-horizon-step lift of Key Lemma 4.* Key Lemma 4 (Lemma 5.4)
 780 applies pointwise at any visited state s_h along a horizon- N_h trajectory: $\mathbb{E}[|\hat{D}_t(s_h) - D_t^{\text{true}}(s_h)|] \leq$
 781 $(1 - p_{\text{id}}(s_h)) \log(1/q_0)$. Aggregating linearly over the N_h horizon steps and assuming the per-state
 782 argmax-correctness floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$, the per-round bias contribution to the trajectory-level
 783 value gap is at most $N_h(1 - p_{\text{id}}) \log(1/q_0)$. Summing over T rounds: $N_h(1 - p_{\text{id}}) \log(1/q_0) \cdot T$.
 784 This is the bias term.

785 *Combining.* Adding Steps 1–4: $\mathbb{E}[\text{DynReg}(T)] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + N_h(1 -$
 786 $p_{\text{id}}) \log(1/q_0) \cdot T$, which is conclusion (v). $\square$

787 **Cesàro tracking corollary** Dividing through by T : $\frac{1}{T} \sum_t (V_t^{\pi_t^*} - V_t^{Q_t}) \leq$
 788 $2c V_{\max} N_h \sqrt{(B_T + 1)/T} + N_h(1 - p_{\text{id}}) \log(1/q_0)$. As $T \rightarrow \infty$ with $B_T = o(T)$ and $p_{\text{id}} \rightarrow 1$, the
 789 right side tends to zero — the agent’s average value over the trajectory tracks the moving optimum’s
 790 average value. $\square$

791 **Bandit case sharpening (Remark)** In the bandit special case ($N_h = 1$, so $\overline{\text{TV}}_t = \mathbb{E}[1 - e^{-K_t}]$),
 792 Thompson sampling and UCB (Section D) satisfy (A5) with the *stronger logarithmic* per-block rate
 793 $\mathbb{E}[1 - e^{-K_t}] = O(\log t / (t \Delta_{\min}))$. Substituting and summing gives cumulative regret $O(V_{\max}(B_T +$
 794 $1) \log^2(T / (B_T + 1)) / \Delta_{\min})$ — sharper than $\sqrt{(B_T + 1)T}$ when $\Delta_{\min}$ is bounded away from zero.
 795 The square-root rate is the worst-case Cauchy–Schwarz bound; the logarithmic rate is the typical-case
 796 bound for stochastic-bandit base learners. For $N_h > 1$ MDPs, UCRL2 and UCBVI provide the

797 natural occupancy-weighted instantiation (Section D) with c of order $\tilde{O}(\sqrt{N_h SA})$, lifting cleanly to
 798 $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ — same piecewise-stationary B_T family as restart-on-change variants of [1].

799 B Proofs of the Key Lemmas

800 This appendix gives full proofs of the four key lemmas surfaced in Section 5, plus the perturbative
 801 extension and the ProST sector-level reduction.

802 B.1 Proof of Key Lemma 1 (Point-mass reverse-KL/TV identity)

803 For deterministic $\pi^* = \delta_{a^*}$ and any policy Q with $Q(a^*) > 0$:

804 The reverse-KL collapses under the point-mass: $D_{\text{KL}}(\delta_{a^*} \| Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)} =$
 805 $\log \frac{1}{Q(a^*)} = -\log Q(a^*)$, where the convention $0 \log 0 = 0$ disposes of the $a \neq a^*$ terms.
 806 Combined with the textbook total-variation calculation $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2} \sum_a |\delta_{a^*}(a) - Q(a)| =$
 807 $\frac{1}{2} (|1 - Q(a^*)| + \sum_{a \neq a^*} Q(a)) = \frac{1}{2} ((1 - Q(a^*)) + (1 - Q(a^*))) = 1 - Q(a^*)$, we have
 808 $-\log Q(a^*) = -\log(1 - \text{TV})$. Solving for TV gives $\text{TV} = 1 - e^{-D_{\text{KL}}}$. $\square$

809 **Strict-improvement substitution** Substituting the identity into the Bretagnolle–Huber inequality
 810 $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$ at $P = \delta_{a^*}$ yields $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$, which holds *strictly*
 811 on $D_{\text{KL}} > 0$ since $x < \sqrt{x}$ on $(0, 1)$. The identity therefore supplies the *exact* TV value at the
 812 deterministic- π^* corner and lies strictly below the BH envelope there.

813 **Two-sided regret bound (Theorem 4.2 in Section 4).** From the regret expression $R(Q) =$
 814 $\sum_{a \neq a^*} Q(a) \Delta(a)$ and $\Delta(a) \leq V_{\max}$: $R(Q) \leq V_{\max} \sum_{a \neq a^*} Q(a) = V_{\max} (1 - Q(a^*)) =$
 815 $V_{\max} \text{TV}(\delta_{a^*}, Q) = V_{\max} (1 - e^{-D_{\text{KL}}})$. The matching action-gap lower bound (full statement in
 816 Section C.4) gives $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min} \text{TV}(\delta_{a^*}, Q) =$
 817 $\Delta_{\min} (1 - e^{-D_{\text{KL}}})$. Combining gives the two-sided characterization. The bound is *Lipschitz-equivalent*
 818 with constants $\Delta_{\min}/V_{\max}$ and 1, both achieved on extremal value landscapes — therefore *coordinate-*
 819 *optimal among bounds depending only on TV*. $\square$

820 B.2 Proof of perturbative extension (Theorem 4.3 — ϵ -stochastic and softmax-regularized)

821 We establish the perturbative identity via a uniform perturbation argument rather than an alignment-
 822 specific calculation.

823 **ϵ -greedy stochastic optimum** Let $\pi_\epsilon^*(a^*) = 1 - \epsilon$ and $\pi_\epsilon^*(a) = \epsilon/(|\mathcal{A}| - 1)$ for $a \neq a^*$, and
 824 assume the full-support lower bound $Q(a) \geq q_0 > 0$ for all $a \in \mathcal{A}$. (Full support is required since π_ϵ^*
 825 has positive mass on every action; without $Q(a) > 0$ on the full support, $D_{\text{KL}}(\pi_\epsilon^* \| Q) = +\infty$.)

826 *Step 1 — TV is Lipschitz under the point-mass perturbation.* $\text{TV}(\pi_\epsilon^*, \delta_{a^*}) = \epsilon$. By the triangle
 827 inequality for total variation, $|\text{TV}(\pi_\epsilon^*, Q) - \text{TV}(\delta_{a^*}, Q)| \leq \text{TV}(\pi_\epsilon^*, \delta_{a^*}) = \epsilon$, uniformly in Q —
 828 no alignment hypothesis needed.

829 *Step 2 — Reverse-KL admits a uniform expansion.* Direct computation:

$$830 \quad D_{\text{KL}}(\pi_\epsilon^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q) = (1 - \epsilon) \log \frac{1 - \epsilon}{Q(a^*)} + \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log \frac{\epsilon/(|\mathcal{A}| - 1)}{Q(a)} - (-\log Q(a^*))$$

$$= (1 - \epsilon) \log(1 - \epsilon) + \epsilon \log Q(a^*) + \epsilon \log \frac{\epsilon}{|\mathcal{A}| - 1} - \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log Q(a),$$

831 where the second equality collects the $\log Q(a^*)$ contributions via $-(1 - \epsilon) \log Q(a^*) + \log Q(a^*) =$
 832 $\epsilon \log Q(a^*)$ and splits the constant log term off the conditional sum. Expanding
 833 $(1 - \epsilon) \log(1 - \epsilon) = -\epsilon + O(\epsilon^2)$ and $\epsilon \log(\epsilon/(|\mathcal{A}| - 1)) = \epsilon \log \epsilon - \epsilon \log(|\mathcal{A}| - 1)$,
 834 the leading-order behavior is $D_{\text{KL}}(\pi_\epsilon^* \| Q) = D_{\text{KL}}(\delta_{a^*} \| Q) + \epsilon \log \epsilon - \epsilon (\log(|\mathcal{A}| - 1) + 1) +$
 835 $\epsilon \log Q(a^*) - \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log Q(a) + O(\epsilon^2)$. Under $Q(a) \geq q_0$, each
 836 $|\log Q(a)| \leq \log(1/q_0)$ is uniformly bounded; the dominant correction is the $\epsilon \log \epsilon$ leading

term, giving $|D_{\text{KL}}(\pi_\epsilon^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q)| = O(\epsilon \log(1/\epsilon) + \epsilon \log(1/q_0)) = O(\epsilon \log(1/\epsilon))$ absorbing $\epsilon \log(1/q_0)$ into the leading $\epsilon \log(1/\epsilon)$ term as $\epsilon \rightarrow 0$ (constants depend on q_0 and $|\mathcal{A}|$).

Step 3 — Map through $1 - e^{-x}$ via Lipschitzness. The map $x \mapsto 1 - e^{-x}$ is 1-Lipschitz on $[0, \infty)$, so $|(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) - (1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)})| = O(\epsilon \log(1/\epsilon))$. Combining Steps 1 and 3 with the unperturbed identity $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)}$: $\text{TV}(\pi_\epsilon^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)} + O(\epsilon \log(1/\epsilon))$ uniformly in Q over the class $\{Q : Q(a) \geq q_0\}$, with constants depending on q_0 and $|\mathcal{A}|$. This establishes the ϵ -greedy form of the perturbative identity. $\square$

Softmax-regularized optimum Let $\pi_\tau^*(a) \propto \exp(Q_O(a)/\tau)$. Under isolated optimum with action gap $\Delta_{\min}$, the off-optimum mass concentrates exponentially in $1/\tau$: $1 - \pi_\tau^*(a^*) = \frac{\sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)}{1 + \sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)} \leq (|\mathcal{A}| - 1) \exp(-\Delta_{\min}/\tau)$. Treating π_τ^* as π_ϵ^* with effective $\epsilon = O(\exp(-\Delta_{\min}/\tau))$, the ϵ -greedy expansion above applies with $\epsilon \log(1/\epsilon) = O((\Delta_{\min}/\tau) \exp(-\Delta_{\min}/\tau)) = O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$, exponentially small with a polynomial prefactor (equivalently, $O(\exp(-c\Delta_{\min}/\tau))$ for any fixed $c \in (0, 1)$ since the exponential dominates the polynomial). $\square$

Two-sided regret bound under perturbation Composing with the TV-regret bounds: $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) - O(\epsilon \log(1/\epsilon)) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) + O(\epsilon \log(1/\epsilon))$.

B.3 Proof of Key Lemma 2 (Multi-factor forgetting prerequisite)

Within Model (S) of Section 5.2 with diagonal correction architecture, the Lyapunov function $V(\delta_\Sigma) := \|\delta_\Sigma\|^2$ satisfies $\mathbb{E}[\Delta V \mid \delta_\Sigma] = -2\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) + \mathbb{E}[\|\mathbf{w}\|^2 \mid \delta_\Sigma]$, where $\mathbf{F}(\delta_\Sigma) := (\alpha_{ij}^{\text{ss}} \delta_{ij})_{(i,j) \in E}$ is the sector correction. The first term is the *sector product*: $\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) = \sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2 \geq \min_{(i,j)} \alpha_{ij}^{\text{ss}} \cdot \|\delta_\Sigma\|^2 = \mathcal{T}_\Sigma^{\text{bn,ss}} \cdot \|\delta_\Sigma\|^2$. The bound is *tight*: setting δ_Σ proportional to the indicator $\mathbf{e}_{(i^*, j^*)}$ where $(i^*, j^*) \in \arg \min_{(i,j)} \alpha_{ij}^{\text{ss}}$ saturates the inequality. The disturbance second-moment $\mathbb{E}[\|\mathbf{w}\|^2] \leq \rho_\Sigma^2$.

Plugging in: $\mathbb{E}[\Delta V \mid \delta_\Sigma] \leq -2\mathcal{T}_\Sigma^{\text{bn,ss}} V + \rho_\Sigma^2$. This is the standard Khalil sector-Lyapunov ultimate-boundedness setup^[28] (Lemma 9.2 / Theorem 9.1 of Chapter 9). Whenever $V > \rho_\Sigma^2 / (2\mathcal{T}_\Sigma^{\text{bn,ss}})$, the expected drift is negative; iterating gives ultimate boundedness with $V \leq R_\Sigma^{*2}$ where $R_\Sigma^{*2} = \rho_\Sigma^2 / \mathcal{T}_\Sigma^{\text{bn,ss}}$ to leading order, i.e., $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$. So whenever $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma / R_\Sigma$, we have $R_\Sigma^* \leq R_\Sigma$ and the modeled mismatch is ultimately bounded within the strategic reserve. $\square$

Sharpness When the inequality reverses ($\mathcal{T}_\Sigma^{\text{bn,ss}} \leq \rho_\Sigma / R_\Sigma$), an adversarial disturbance concentrating on the bottleneck element (i^*, j^*) with $|w_{i^* j^*}| = \rho_\Sigma$ pushes $\|\delta_\Sigma\| > R_\Sigma$ within the modeled dynamics — the threshold is sharp inside the diagonal sector model. The theorem is silent about non-diagonal correction architectures or stabilization mechanisms outside Model (S). $\square$

B.4 Proof of impulsive ProST reduction (Lemma 5.2)

Within Model (S), idealize ProST^[8] as an impulsive system: between scheduled update times $\{t_1, \dots, t_K\} \subset [0, T]$ the agent's policy is held fixed and the strategic mismatch evolves under disturbance budget ρ_Σ alone (continuous-destabilizing in the Lyapunov $V = \|\delta_\Sigma\|^2$); at each update time t_i the policy is revised, contracting the modeled mismatch by per-update impulse gain $\gamma \in (0, 1]$ so $\|\delta_\Sigma(t_i^+)\| \leq (1 - \gamma)\|\delta_\Sigma(t_i^-)\|$. Then $V(t_i^+) \leq (1 - \gamma)^2 V(t_i^-)$.

This fits the Hespanha–Liberzon–Teel^[46] (Theorem 1) framework for impulsive systems with destabilizing continuous evolution and stabilizing impulses (the *reverse-ADT* branch). The continuous-evolution rate is $\lambda_c = \rho_\Sigma / R_\Sigma$ (from linearizing the disturbance contribution to $\dot{V} \approx +\lambda_c V + \gamma_c(\rho_\Sigma)$). The impulse contraction in V is $\mu = (1 - \gamma)^2$. The reverse-ADT condition for ultimate boundedness — impulses cannot be too sparse — gives $\Delta_{\max} \cdot \lambda_c < -\ln \mu \iff \Delta_{\max} \cdot \rho_\Sigma / R_\Sigma < -\ln(1 - \gamma)^2$, where $\Delta_{\max} := \sup_i (t_{i+1} - t_i)$ is the longest inter-update gap. Under the condition the modeled mismatch is ultimately bounded within $R_\Sigma^* \approx \rho_\Sigma \Delta_{\max} / \gamma$ (leading order in γ).

Under uniform blocks $\Delta_i = T/K$ the condition reduces to $K/T > \rho_\Sigma/(2\gamma R_\Sigma)$ to leading order in γ — recovering ProST’s K/T form with the impulse gain made explicit. Under nonuniform schedules the impulsive form is *strictly stronger*: the longest block sets the threshold, not the average. The lemma is rigorous for the modeled impulsive sector dynamics; the per-update impulse gain γ is a domain quantity (depends on within-block sample size and update-time discount) and is left abstract. $\square$

B.5 Proof of Key Lemma 3 (Loop generates interventional samples)

By temporal ordering and the singular-trajectory commitment of Section 3, a_t causally precedes o_{t+1} . Under (C1) positivity, (C2) sequential ignorability, and (C3) known action mechanism, conditioning on the information state H_t blocks the action-selection confounding pathway. Specifically, (C2) gives $a_t \perp\!\!\!\perp (\text{environment latents}) \mid H_t$, so for any subset of environment-latent variables $\mathbf{U}$, $P(o_{t+1} \mid a_t, H_t, \mathbf{U}) = P(o_{t+1} \mid a_t, H_t)$, on the support where $\pi_t(a_t \mid H_t) > 0$ (which (C1) guarantees). The interventional distribution on the same support is, by Pearl’s do-calculus^[35], $P(o_{t+1} \mid \text{do}(a_t), H_t) = \sum_{\mathbf{U}} P(o_{t+1} \mid a_t, H_t, \mathbf{U}) \cdot P(\mathbf{U} \mid H_t) = \sum_{\mathbf{U}} P(o_{t+1} \mid a_t, H_t) \cdot P(\mathbf{U} \mid H_t) = P(o_{t+1} \mid a_t, H_t)$, where the second equality uses the conditional-independence factorization above. So the conditional and interventional distributions coincide on the support. (C3) makes the agent’s action-mechanism identification well-defined; the architecture of the agent does not enter — the loop generates conditional-Level-2 samples whenever (C1)–(C3) hold. $\square$

B.6 Proof of Key Lemma 4 (Bias bound)

On the matching event $\{a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)\}$, the difference $\hat{D}_t(s) - D_t^{\text{true}}(s)$ vanishes identically (both quantities equal $-\log Q_t$ at the same action). On the complementary misidentification event, the support condition $Q_t(\cdot \mid s) \geq q_0$ at *both* $a_{\text{ag},t}^*(s)$ and $\tilde{a}_t^*(s)$ implies $\log Q_t(a_{\text{ag},t}^*(s) \mid s) \in [\log q_0, 0]$ and $\log Q_t(\tilde{a}_t^*(s) \mid s) \in [\log q_0, 0]$, so their absolute difference is at most $\log(1/q_0)$. Therefore $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$. Taking expectations: $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$ by definition of $p_{\text{id}}(s)$. $\square$

High-probability form With probability $1 - \delta$ over a single round’s identification event, $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \log(1/q_0) \cdot \mathbf{1}[\text{misdevent}]$ where $\Pr[\text{misdevent}] = 1 - p_{\text{id}}(s)$. The bias decomposes cleanly: in Regime A ($p_{\text{id}}(s) \rightarrow 1$) it vanishes; in Regime B ($p_{\text{id}}(s) \in (0, 1)$) it is controlled by misidentification mass; in Regime C ($p_{\text{id}}(s) \rightarrow 0$) it saturates at $\log(1/q_0)$. This proves Key Lemma 4. $\square$

Per-state, per-horizon-step lift (used in Theorem 4.1(v)). The lemma applies pointwise at any visited state s_h along a horizon- N_h trajectory: $\mathbb{E}[|\hat{D}_t(s_h) - D_t^{\text{true}}(s_h)|] \leq (1 - p_{\text{id}}(s_h)) \log(1/q_0)$, with $p_{\text{id}}(s_h)$ the per-state argmax-correctness probability. Aggregating linearly over the horizon and assuming the per-state floor $p_{\text{id}} := \min_s p_{\text{id}}(s)$, the per-round bias contribution to the trajectory-level value gap is at most $N_h(1 - p_{\text{id}}) \log(1/q_0)$, contributing the second term of Theorem 4.1(v).

Remarks - Both $a_{\text{ag},t}^*(s)$ and $\tilde{a}_t^*(s)$ must satisfy $Q_t \geq q_0$ for the $\log(1/q_0)$ ceiling to apply; the support condition is on *both* points, not just the agent’s estimate. This is *strictly weaker* than the perturbative identity’s full-support condition $Q(a) \geq q_0$ for *all* a (Section B.2) — the perturbative identity needs full support because π_ϵ^* has positive mass on every action; the bias bound here needs only two-point support. - This is *not* an estimation-from-samples lemma. If Q_t is unavailable internally and must be estimated from on-policy rollouts, an additional concentration argument (Hoeffding under iid fixed-policy samples, or martingale concentration for adaptive policies) is needed; in the architectures this paper considers, Q_t is the agent’s policy and is internally available.

C Auxiliary Mathematical Material

This appendix collects supporting material used by the main results: convention hierarchy details, the admissible-divergence family, direction-forcing argument, action-gap matching lower bound, tied-optimum extensions, the structural-class theorem on gain-decay updates, strategic-tempo topologies, deployment modes, and the chain-rule uniqueness theorem.

930 C.1 Convention hierarchy: C1, C2, C3

931 Three named continuation conventions form a hierarchy of increasing diagnostic power and computa-
932 tional cost.

933 **C1 — One-step improvement (canonical default)** $\pi_{\text{cont}} = \pi_{\text{current}}$. Each action is evaluated
934 assuming current behavior continues afterward. No fixed-point computation, no global-optimality
935 assumption. Cheapest; weakest diagnostic.

936 **C2 — Receding-horizon (N_r -step replanning)** At each future step, re-optimize
937 over a horizon of N_r steps using the model available at that step: $\pi_{\text{RH}}(M_\tau) =$
938 $\arg \max_\pi V_O(M_\tau, \pi; N_r)$ applied at each τ . Captures multi-step recovery: a goal that ap-
939 pears unattainable under frozen continuation may be reachable with replanning. Moderate cost;
940 moderate diagnostic.

941 **C3 — Bellman** $\pi_{\text{cont}} = \pi^*$ where $\pi^* = \arg \max_\pi V_O(M_t, \pi; N_h)$. The continuation is the optimal
942 policy — a fixed-point equation. Strongest diagnostic; most expensive.

943 **Proposition C.1 (Monotonicity).** *For any model M_t , horizon N_h , and policy class Π :*
944 $A_O^{(1)}(M_t; \Pi, N_h) \leq A_O^{\text{RH}}(M_t; \Pi, N_r, N_h) \leq A_O^{\text{B}}(M_t; \Pi, N_h)$.

945 *Proof.* C1 freezes continuation at π_{current} (possibly suboptimal); C2 re-optimizes periodically, so
946 $\pi_{\text{RH}} \succeq \pi_{\text{current}}$ at each future step; C3 uses the globally optimal π^* . A weakly better continuation
947 yields a weakly higher expected trajectory value; taking the supremum over the first action preserves
948 the ordering. $\square$

949 **Corollary** $\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)}$ (since $\delta_{\text{sat}} = V_O^{\min} - A_O$, higher A_O means lower δ_{sat}). And
950 $\delta_{\text{regret}}^{(1)} \leq \delta_{\text{regret}}^{\text{RH}} \leq \delta_{\text{regret}}^{\text{B}}$ (since $\delta_{\text{regret}} = A_O - V_O(M_t, \pi_{\text{current}}; N_h)$, higher A_O means higher δ_{regret}).

951 C1 is the most conservative diagnostic (most likely to diagnose “locally unattainable”); C3 is the most
952 accurate (least false “unattainable” diagnoses). The 2×2 diagnostic structure is preserved under all
953 three; only the inferential force varies.

954 C.2 Admissible-divergence family for the regret bound

955 The reverse-KL direction is forced by the deterministic- π^* regime (Section C.3). Within the direction-
956 forced family, multiple smooth f -divergences yield valid regret bounds:

	Divergence	Bound on $R(Q)$	Tightness	Finite under det. π^* ?
	$\text{TV}(\pi^*, Q)$	$V_{\max} \cdot \text{TV}$	Tight (extremal V)	Yes
	$D_{\text{KL}}(\pi^* \parallel Q)$ via Pinsker	$V_{\max} \sqrt{D_{\text{KL}}/2}$	Loose by $\sqrt{\cdot}$	Yes
957	$D_{\text{KL}}(\pi^* \parallel Q)$ via point-mass identity	$V_{\max}(1 - e^{-D_{\text{KL}}})$	Tight (this paper)	Yes
	$\chi^2(\pi^* \parallel Q)$ (Le Cam)	$V_{\max} \cdot \frac{1}{2} \sqrt{\chi^2}$	Looser than Pinsker	$\chi^2 = 1/Q(a^*) - 1$
	$D_\alpha(\pi^* \parallel Q)$ (Rényi, $\alpha \geq 1$)	Various	Interpolates	Yes for $\alpha \geq 1$
	$D_{\text{KL}}(Q \parallel \pi^*)$ (forward-KL)	$+\infty$	Vacuous	No

958 Reverse-KL is selected uniquely within the direction-forced family by the chain-rule axiom (Sec-
959 tion C.10). The point-mass identity supplies the *exact* bound on reverse-KL under deterministic π^* ; the
960 table reflects different bound shapes on the same divergence (with the BH inequality $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$
961 as the looser general form to which the identity reduces outside the deterministic- π^* corner).

962 C.3 Direction-forcing argument

963 For deterministic $\pi^* = \delta_{a^*}$ and any Q with $Q(a) > 0$ for some $a \neq a^*$: $D_{\text{KL}}(Q \parallel \pi^*) =$
964 $\sum_a Q(a) \log \frac{Q(a)}{\pi^*(a)} = \sum_{a \neq a^*} Q(a) \log \frac{Q(a)}{0} = +\infty$. A bound “ $R \leq +\infty$ ” is vacuous. The
965 reverse direction $D_{\text{KL}}(\pi^* \parallel Q)$ is finite (and equal to $-\log Q(a^*)$) whenever $Q(a^*) > 0$. The asym-
966 metry is forced by the regime: regret is a one-sided quantity (contributes only from Q ’s off-optimum
967 mass; π^* has no support off a^*); divergences whose role is to bound this one-sided quantity must

968 themselves be one-sided. Symmetric divergences (squared Hellinger, Jensen-Shannon, symmetrized
969 KL) introduce a vacuous symmetric term.

970 C.4 Action-gap matching lower bound

971 For any Q with $\Delta_{\min} = \min_{a \neq a^*} \Delta(a) > 0$: $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) =$
972 $\Delta_{\min} \cdot (1 - Q(a^*)) = \Delta_{\min} \cdot \text{TV}(\pi^*, Q)$. By the point-mass identity (Lemma 5.1), $\text{TV}(\pi^*, Q) =$
973 $1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$, giving the matching lower bound used in the two-sided characterization.

974 The lower bound is tight when sub-optimal actions are uniformly bad ($\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$). For
975 typical landscapes the gap between upper and lower bound is $V_{\max} - \Delta_{\min}$, controlled by the *spread*
976 of action gaps.

977 C.5 Tied-optimum and softmax-smoothed extensions

978 **Tied-optimum** If π^* has support on a tied-optimum set $\mathcal{A}^* = \{a : Q_O(a) = Q_O(a^*)\}$
979 with uniform mass, reverse-KL is finite whenever Q covers $\mathcal{A}^*$. The regret bound extends:
980 $R(Q) = \sum_{a \notin \mathcal{A}^*} Q(a) \Delta(a) \leq V_{\max} \cdot \mathbb{P}_Q(a \notin \mathcal{A}^*)$. A multi-action analog of the point-mass
981 identity holds with $\pi^*(a^*) = 1/|\mathcal{A}^*|$ replacing the point-mass form, yielding $D_{\text{KL}}(\pi^* \| Q) =$
982 $-\sum_{a \in \mathcal{A}^*} (1/|\mathcal{A}^*|) \log(|\mathcal{A}^*| Q(a))$, a multi-action analog of $-\log Q(a^*)$. The two-sided bound be-
983 comes one-sided in general; the upper bound holds with the modified KL form.

984 **Softmax-smoothed π^* with temperature $\tau \rightarrow 0$** Replace $\pi^* = \delta_{a^*}$ with $\pi_\tau^* \propto \exp(Q_O/\tau)$. As
985 $\tau \rightarrow 0$, $\pi_\tau^* \rightarrow \delta_{a^*}$. For finite τ , the *perturbative* identity (Section B.2) applies: $\text{TV}(\pi_\tau^*, Q) =$
986 $1 - e^{-D_{\text{KL}}(\pi_\tau^* \| Q)} + O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ — exponentially small with polynomial prefactor. The
987 two-sided regret bound transfers with the same correction order. Outside the perturbative regime —
988 for genuinely high-entropy optima — the BH inequality $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ becomes the relevant
989 general bound; Pinsker also applies, with a one-sided regret form.

990 C.6 The gain-decay structural class $\mathcal{A}_{\text{decay}}$

991 The persistence inequality $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$ (Section 5.2) is an *instantaneous* check at the current
992 operating point. Define the structural class by gain-decay rather than sample-counting: $\mathcal{A}_{\text{decay}} :=$
993 $\{\text{updates whose effective sector gain } \alpha_{ij}^{(t)} \rightarrow 0 \text{ as } t \rightarrow \infty \text{ on every revisable element}\}$.

994 **Theorem C.2** (Universal failure of $\mathcal{A}_{\text{decay}}$). *For any fixed $(\rho_{\Sigma}, R_{\Sigma})$ with $\rho_{\Sigma} > 0$, every agent in*
995 *$\mathcal{A}_{\text{decay}}$ eventually violates the persistence threshold $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$.*

996 *Proof.* At every element (i, j) , $\alpha_{ij}^{(t)} \rightarrow 0$ by definition of $\mathcal{A}_{\text{decay}}$, so $\min_{(i,j)} \alpha_{ij}^{(t)} \rightarrow 0$, so the
997 bottleneck $\mathcal{T}_{\Sigma}^{\text{bn}} \rightarrow 0$ with experience. Once $\mathcal{T}_{\Sigma}^{\text{bn,ss}} < \rho_{\Sigma}/R_{\Sigma}$, the threshold is violated. This holds
998 *regardless of prior calibration*: at any finite calibration level, the gain decays below threshold once
999 enough experience accumulates. $\square$

1000 This is a *structural* failure of the class, not a tuning problem. The class includes: - Count-accumulating
1001 Bayesian updates without forgetting (e.g., Beta-Bernoulli with $\eta_{\text{edge},ij} = 1/(n_{ij}+1)$ where $n_{ij} \rightarrow \infty$).
1002 - Bounded-memory schemes with growing memory. - Observation-aggregating schemes without restart.
1003 - Gradient-based methods with vanishing step sizes ($\eta_t = c/t^\beta$ for $\beta \in (0, 1]$, the Robbins–Monro
1004 regime).

1005 Mechanisms *outside* $\mathcal{A}_{\text{decay}}$ — constant-step-size stochastic approximation, sliding-window updates,
1006 bounded-memory learners, block-restart schemes, Kalman filters with bounded process noise —
1007 maintain a finite gain ceiling and escape asymptotic decay, but then face a *bidirectional* threshold:

Mechanism	$\alpha_{\Sigma}^{\text{ss}}$
Exponential forgetting with λ	$1 - \lambda$
Constant-step SA with rate η	η
Fixed-window mechanisms (sliding-window size W , bounded-memory size K , block-restart period T_R)	$1/W, 1/K, 1/T_R$

1009 For each row, the threshold is *bidirectional* and sharp within the sector-Lyapunov reduction.

1010 C.7 Strategic-tempo across canonical topologies

1011 The bottleneck strategic tempo $\mathcal{T}_{\Sigma}^{\text{bn,ss}}$ of Lemma 5.2 is verified — and the resulting per-topology
1012 threshold derived — across four canonical topologies under Beta-Bernoulli edge updates with per-
1013 element forgetting:

- 1014 • **(T1) Single edge.** $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \nu \cdot \iota \cdot (1 - \lambda)$. All three factors load-bearing; setting any one to
1015 zero collapses the bottleneck.
- 1016 • **(T2) Two-edge AND chain, observable intermediate.** $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min(\nu_1(1 - \lambda_1), \nu_1\theta_1(1 -$
1017 $\lambda_2))$. Edge 2’s effective observation rate is gated by edge 1’s success probability θ_1 —
1018 *depth-gated attenuation*. For depth- d chains, the bottleneck is $\min_k \prod_{j < k} \theta_j(1 - \lambda_k)$.
- 1019 • **(T3) Two-edge AND chain, unobservable intermediate.** Per-edge tempo ill-defined; plan-
1020 level bottleneck $\mathcal{T}_{\Sigma, \text{plan}}^{\text{bn,ss}} = \nu(1 - \lambda_{\Phi})$ over a single tracked plan-level quantity $\hat{\Phi} = p_1 p_2$.
- 1021 • **(T4) Two-arm OR node, ε -greedy.** $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min((1 - \varepsilon)(1 - \lambda_1), \varepsilon(1 - \lambda_2))$. Action
1022 selection controls rate allocation; pure greedy ($\varepsilon = 0$) collapses the unexplored arm’s
1023 bottleneck — *exploration-gated*.

1024 The structural decomposition: AND-chains exhibit *depth-gated* geometric attenuation $\nu_k =$
1025 $\nu \prod_{j < k} \theta_j$; OR-nodes exhibit *exploration-gated* allocation. Mixed AND/OR DAGs interleave both.
1026 Each topology surfaces a different factor as the binding bottleneck.

1027 **Regime-C contamination is structurally fatal** $\iota_{ij} = 0$ at any single element $\Rightarrow \mathcal{T}_{\Sigma}^{\text{bn,ss}} = 0$. An
1028 agent with even one fully-confounded element cannot meet the threshold by forgetting alone.

1029 C.8 Three deployment modes of loop-Level-2

1030 The Pearl-do structure of closed-loop interventional access manifests at distinct layers with semanti-
1031 cally distinct interventional mechanisms:

- 1032 • **Mode 1 — agent-self-intervention.** The agent performs do-actions on its own action space
1033 as part of its ordinary adaptive loop. Intervention is on the agent’s own action; target is the
1034 environment’s response. This is Lemma 5.3’s primary content.
- 1035 • **Mode 2 — observer-on-sub-agent.** An observer external to a composite performs do-
1036 interventions on one sub-agent’s action space; target is another sub-agent’s response. Reveals
1037 cross-coupling structure that component-marginal observation cannot identify.
- 1038 • **Mode 3 — observer-on-agent-input** (sketched, future work). An observer intervenes on
1039 the agent’s observation channel; target is the agent’s subsequent policy. Useful for offline
1040 architectural-class diagnosis.

1041 Modes share the Pearl-do structure but differ in who intervenes on what. The unification is at the
1042 pattern level; the mechanism is semantically distinct per layer.

1043 C.9 Distinction from active inference and causal-RL precursors

1044 Action-perception-loop frameworks — active inference^[30,31], control-as-inference^[48], cybernet-
1045 ics^[49,50] — implicitly use the action-causes-observation observation. Our distinctive moves:

- 1046 • *Bareinboim-hierarchy connection.* Active inference / cybernetics rest on Bayesian-network
1047 (Level 1) generative models; we invoke the causal-hierarchy theorem^[29] to position the
1048 policy DAG as causal with do-conditioning in Q_O .
- 1049 • *Regime-indexed identifiability (A/B/C).* Active inference literature treats causal identifiability
1050 uniformly within modeling assumptions; we surface the regime split at framework level.
- 1051 • *Scope honesty.* We distinguish “data generated under intervention” from “cleanly identified
1052 do-estimates”;^[47] documents that the active-inference literature sometimes elides this.

1053 The causal-RL line^[13–18] is the direct ancestor for regime-indexed identifiability and on-policy inter-
 1054 ventional access; all are stationary-MDP. Composition with non-stationarity is, to our knowledge,
 1055 novel.

1056 C.10 Chain-rule uniqueness of reverse-KL

1057 **Theorem C.3** (Chain-rule uniqueness (Hobson 1969; Csiszár 1991)). *Let $D_f(P \parallel Q) =$
 1058 $\sum_x Q(x) f(P(x)/Q(x))$ be a smooth f -divergence with f convex and $f(1) = 0$. The chain rule
 1059 $D_f(P_{XY} \parallel Q_{XY}) = D_f(P_X \parallel Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \parallel Q_{Y|X})]$ holds for all joint distributions
 1060 if and only if $f(t) = c \cdot t \log t$ for some $c > 0$ — i.e., D_f is reverse-KL up to positive scaling.*

1061 *Proof.* Writing $r_x = P(x)/Q(x)$ and $s_{y|x} = P(y|x)/Q(y|x)$, the chain rule reduces to the functional
 1062 equation $f(rs) = f(r) + rf(s)$ for all $r, s > 0$. With $f(1) = 0$ and convexity, the unique solution is
 1063 $f(t) = c \cdot t \log t$ for $c > 0$ ^[51] (§4). $\square$

1064 **References** Hobson 1969 (“A new theorem of information theory,” *J. Stat. Phys.*); Csiszár 1991
 1065 (“Why least squares and maximum entropy?” *Annals of Statistics*; Theorem 3 corollary, Theorem 5);
 1066 Shore-Johnson 1980 (“Axiomatic derivation of the principle of maximum entropy,” *IEEE Trans. Info.*
 1067 *Theory*, system-independence axiom); Sanov 1957 (large-deviation rate function); Aczél-Daróczy
 1068 1975 (functional-equation machinery).

1069 These references give *structurally equivalent reformulations* of the same axiom. The Cauchy functional
 1070 equation each reduces to is the common content. No known uniqueness route outside the independence-
 1071 on-sub-problems family exists.

1072 **Why other family members fail the chain rule** Concrete counterexample for χ^2 : take Q_X uniform
 1073 on $\{x_1, x_2\}$, $P_X = (3/4, 1/4)$, $Q(y|x)$ uniform, $P(y|x) = (3/4, 1/4)$. Direct calculation gives
 1074 $\chi^2(P_{XY} \parallel Q_{XY}) = 9/16$ while $\chi^2(P_X \parallel Q_X) + \mathbb{E}_{P_X}[\chi^2(P_{Y|X} \parallel Q_{Y|X})] = 1/4 + 1/4 = 8/16$.
 1075 Non-additive. Rényi- α for $\alpha \neq 1$ fails analogously; squared Hellinger likewise fails.

1076 D Algorithm Sketch and Base-Learner Instantiation

1077 A practical algorithm achieving the joint guarantees of Theorem 4.1 requires four components, each
 1078 translating one of the assumptions (A1)–(A5) into operational form.

1079 **(a) Identity-tight base learner satisfying (A5)** In the stationary deterministic- π^* bandit case ($N_h =$
 1080 1), Thompson sampling^[19] and UCB^[27] both achieve the per-round coordinate $\mathbb{E}[1 - e^{-K_t}] =$
 1081 $\mathbb{E}[1 - Q_t(a^*)] = O\left(\frac{\log t}{t \Delta_{\min}}\right)$, matching the identity form $V_{\max}(1 - e^{-D_{\text{KL}}})$ of Lemma 5.1 up
 1082 to constants and a $\log t$ factor. This satisfies (A5) of Theorem 4.1 with the *stronger logarithmic*
 1083 *rate*, yielding cumulative dynamic regret $O(V_{\max}(B_T + 1) \log^2(T/(B_T + 1))/\Delta_{\min})$. For $N_h > 1$
 1084 MDPs, UCRL2^[52] and UCBVI^[43] give per-block cumulative trajectory-level regret $\tilde{O}(N_h^{3/2} \sqrt{SAL})$
 1085 with S, A the state/action sizes; rewriting in the occupancy-weighted coordinate satisfies (A5) with
 1086 c of order $\tilde{O}(\sqrt{N_h SA})$, lifting cleanly to the $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$ cumulative rate — same
 1087 piecewise-stationary B_T family as restart-on-change variants of^[11] (the continuous-variation regime
 1088 of^[3] is structurally distinct; see Section 4 *Comparator regime*). The occupancy-weighted form is the
 1089 natural object for trajectory-level base learners; the per-state KL/TV identity is preserved underneath.
 1090 Information-directed sampling^[20] requires a different analysis here because $H(\pi^*) = 0$ collapses its
 1091 information ratio at the deterministic- π^* corner; we leave the IDS analysis as future work.

1092 **(b) Multi-factor forgetting schedule satisfying (A2)** Choose per-element discount rates $\{\lambda_{ij}\}$
 1093 such that the bottleneck $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min_{(i,j)} \nu_{ij} \iota_{ij} (1 - \lambda_{ij})$ exceeds ρ_{Σ}/R_{Σ} . When ρ_{Σ} is unknown,
 1094 the^[2] black-box reduction gives an adaptive choice (applied per element if heterogeneous); when ρ_{Σ}
 1095 is known via a variation-budget^[11] or change-point detection^[39], the choice is direct. As a fail-fast
 1096 pre-check, $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$ is necessary.

1097 **(c) Identifiability check satisfying (A3)** Test whether the loop data is in Regime A (full identifiability), Regime B (partial), or Regime C (none). In Regime A, no adjustment is needed. In Regime B, apply^[17] (DOVI)-style confounding-bias adjustment. In Regime C, the bound is provable but not realizable; algorithm flags the regime as out-of-scope.

1101 **(d) Two-gap diagnostic readout satisfying (A4)** Compute δ_{sat} (requires A_O , intractable in general; estimable via the policy-class supremum over recent rounds) and δ_{regret} (requires V_O at current policy; tractable in simulation). Route to corrective action class.

1104 The algorithm is sketch-grade; full instantiation and empirical evaluation are deferred to a follow-up paper.

1106 E Pinsker vs. Point-Mass Identity Numerical Comparison

1107 We compare $V_{\max}(1 - e^{-D_{\text{KL}}})$ (point-mass identity bound, exact under deterministic π^* per Lemma 5.1 and the two-sided regret theorem) with $V_{\max}\sqrt{D_{\text{KL}}/2}$ (Pinsker bound, also valid under deterministic π^* but loose), $V_{\max}\sqrt{1 - e^{-D_{\text{KL}}}}$ (Bretagnolle–Huber inequality, the general envelope below which the identity sits at this corner), and the trivial envelope $V_{\max}$. Set $V_{\max} = 1$ for normalization.

	D_{KL}	$1 - e^{-D_{\text{KL}}}$ (identity)	$\sqrt{1 - e^{-D_{\text{KL}}}}$ (BH)	$\sqrt{D_{\text{KL}}/2}$ (Pinsker)	$\min(\sqrt{D_{\text{KL}}/2}, 1)$	Pinsker / identity ratio
	0.01	0.00995	0.0998	0.0707	0.0707	7.10×
	0.1	0.0952	0.308	0.224	0.224	2.35×
1111	0.5	0.393	0.627	0.500	0.500	1.27×
	1.0	0.632	0.795	0.707	0.707	1.12×
	2.0	0.865	0.930	1.000	1.000	1.16×
	4.0	0.982	0.991	1.414	1.000	1.02×
	10.0	0.99995	0.99998	2.236	1.000	1.00×

1112 The point-mass identity bound is uniformly tighter than Pinsker, by a factor of $7\times$ at $D_{\text{KL}} = 0.01$ and converging to 1 as D_{KL} grows large (where both saturate at $V_{\max}$). It is also uniformly tighter than the BH inequality $\sqrt{1 - e^{-D_{\text{KL}}}}$ — the relation $x < \sqrt{x}$ on $(0, 1)$ — although both saturate together at $V_{\max}$ for large D_{KL} , where BH stays informative while Pinsker becomes vacuous (exceeds the trivial $V_{\max}$ envelope) for $D_{\text{KL}} > 2$.

1117 The matching lower bound $\Delta_{\min}(1 - e^{-D_{\text{KL}}})$ has the same shape, scaled by $\Delta_{\min}/V_{\max}$. In the extremal value landscape ($\Delta_{\min} = V_{\max}$), the upper and lower bounds coincide and the regret is *identified exactly* by the KL coordinate. For typical landscapes, regret and KL are Lipschitz-equivalent with constants $\Delta_{\min}/V_{\max}$ and 1.

1121 F Prior-Art Retrieval Summary

1122 We conducted a structured prior-art retrieval (via Undermind) with the goal of identifying any framework composing all four target elements: (1) goal-feasibility-vs-policy-quality two-gap diagnostic; (2) an exact reverse-KL/TV regret identity (point-mass) under deterministic- π^* , or — more loosely — Bretagnolle–Huber-style use in regret analysis; (3) strategic-tempo / forgetting prerequisite tying convergence to rate of useful policy revision; (4) closed-loop interventional access making regret bounds learnable from on-policy data.

1128 **Result: 63 papers retrieved, abstract-level coverage estimated 75%, no direct anticipation in the retrieved set** The landscape splits into four largely separate strands corresponding to our four components; we did not find a published framework composing all four in this retrieval. Retrieval queries used the structured prior-art tool Undermind; the four-strand cite-and-distinguish in Section 2 surfaces the closest neighbor in each strand.

1133 **Strongest negative signal (in the retrieved corpus): the Bretagnolle–Huber inequality is absent as an upper-bound coordinate for regret analysis in the retrieved RL / non-stationary literature** The information-theoretic regret line —^[19–24] — uses entropy / mutual information / information ratio / Pinsker / Hellinger; we did not find the BH inequality deployed *as an upper-bound regret*

1137 *coordinate*. (BH is well-known in bandit *lower-bound* proofs via Le Cam’s method and Tsybakov’s
1138 framework^[26], including in standard RL textbook treatments^[27]; that use is not contested here.) The
1139 *exact point-mass identity at the deterministic- π corner** (Lemma 5.1) — which strictly improves both
1140 Pinsker and the BH envelope at this corner — was likewise not found in the retrieved corpus, in either
1141 upper-bound or lower-bound use. We frame this as a literature-search observation against an explicitly
1142 bounded retrieval (63 papers, ~75% abstract-level coverage), not a formal absence claim against all
1143 published RL work; reviewers should read it as “to our knowledge / in the retrieved set” rather than
1144 “absolutely novel.”

1145 **Closest neighbors per strand:**^[1] (Strand 1 — variation-budget dynamic regret);^[5] (Strand 2 —
1146 closest two-term decomposition, different axis);^[8] (Strand 3 — closest tempo result, different form);^[14]
1147 (Strand 4 — closest causal-RL, stationary only).

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